Pressure Equipment Directive 2014/68/EU(97/23/EC) Training

AGENDA

- Introduction to principles behind EU Product Directives (New Approach) / CE Marking
- Introduction to PED - Scope and Purpose
- Annex II – Design & Categorization
- Materials
- Permanent Joining (Welding) and NDE
- Final Assessment
- Annex I – ESRs
- Annex III - Conformity Assessment Modules
- PED in Practice
- Questions and Answers

EU Product Directive

EU Product Directive

- ✓ A Law passed by the EU that must be implemented by all EEA member states. The law (Directive) is binding upon the member state as to the objective to be achieved, with limited leeway as to how to achieve it.
- ✓ Intended to remove technical barriers to trade by establishing a commonly accepted level of **safety**.
- ✓ Example: Directive 2014/68/EU(97/23/EC) covering **Pressurised Equipment**

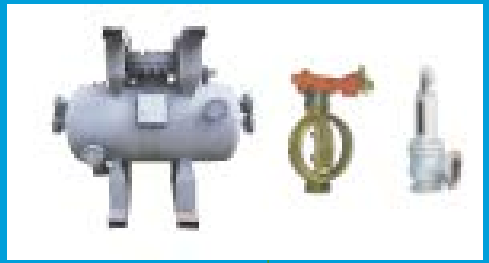
EU Product Directive

EMC Directive

https://en.wikipedia.org/wiki/List_of_European_Union_directives

Machinery Directive

Low Voltage Directive



Pressure Equipment Directive

ATEX Directive

Definitions

✓Placing on the market

The making available, against payment or free of charge, of a product covered by the Directive, and intended for a user, in the Community market for the purpose of distribution and/or use on the Community territory

✓Putting into service

The first use on the Community territory, by its end user, of a product referred to in the Directive

Single market Trade

Definitions, contd.

✓Manufacturer

Any natural or legal person who designs or manufactures a product or who has such a product designed or manufactured, and markets that product under his name or trademark

✓Authorised representative

Any natural or legal person established within the Community, appointed by the manufacturer by a written mandate, to act on his behalf in respect of certain obligations of the manufacturer

✓Importer

Any natural or legal person established within the Community, who places a product from a third country on the Community market

Ref.: http://ec.europa.eu/enterprise/regulation/internal_market_package/docs/com_2007_0053_en.pdf

EEA Member States Enforcing EU Product Directives:

The European Commission promotes the global and sustainable competitiveness of the sectors within the Pressure Equipment and Gas Industries by assessing the challenges and taking necessary actions. The Commission also coordinates and monitors the implementation and interpretation of a number of important **Directives related to Pressure Equipment and Gas Appliances** . It also oversees standardization for these sectors.

https://europa.eu/european-union/about-eu/countries_en

Country List of EU members

EEA Member States Enforcing EU Product Directives:

European Union Members (EU: 28MS)

Germany, France, UK, Italy, Netherlands, Spain, Denmark, Belgium, Ireland, Portugal, Greece, Luxembourg, Austria, Finland, Sweden, Czech Republic, Cyprus, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia, Slovenia, Bulgaria and Romania

European Free Trade Association (EFTA: 4MS)

Norway, Iceland, Liechtenstein, Switzerland

European Economic Area (EEA)

EU+EFTA (-Switzerland)

However Switzerland is linked to the European Union by the Swiss–EU bilateral agreements.

You are an **Economic operator** under the PED if you are:

- ✓A manufacturer
- ✓An authorized representative for a manufacturer located outside of the European Union
- ✓An importer
- ✓A distributor
- ✓The revision also makes clear that if an importer or distributor places pressure equipment or assemblies on the market under their own names or trademarks, or modifies the equipment in a certain way, they will be considered manufacturers and will therefore have to take legal responsibility for the product's compliance including conformity assessment . Future articles will examine in more detail the roles and responsibilities of 'economic operators', the NLF (NEW LEGAL FRAMEWORK)concepts of 'placing on the market' and 'making available on the market' and the implications for economic operators including how the transition from 97/23/EC to 2014/68/EU will affect you.

Manufacturer Responsibility in Brief

- ✓If you are a manufacturer, you have to follow these six steps to affix a CE marking to your product:
- ✓Identify the applicable directive(s) and harmonized standards
- ✓Verify product specific requirements
- ✓Identify whether an independent conformity assessment (by a notified body) is necessary
- ✓Test the product and check its conformity
- ✓Draw up and keep available the required technical documentation
- ✓Affix the CE-marking and draw up the EU Declaration of Conformity.

NOTIFIED BODY

“ Notification is an act whereby a Member State informs the Commission and the other Member States that a body, which fulfils the relevant requirements, has been designated to carry out conformity assessment according to a directive. Notification of Notified Bodies and their withdrawal are the responsibility of the notifying Member State ”

(http://ec.europa.eu/enterprise/newapproach/nando/index.cfm?fuseaction=directive.main)

NB 0427	DET NORSKE VERITAS B.V. - PCNL	Netherlands
NB 0097	DET NORSKE VERITAS BV (DNV)	United Kingdom
NB 0496	DET NORSKE VERITAS Italia Srl	Italy
NB 1347	DET NORSKE VERITAS, DANMARK A/S	Denmark
NB 0575	DNV GL AS	Norway
NB 0434	DNV GL Business Assurance Norway AS	Norway
NB 1162	DNV GL Business Assurance Sweden AB	Sweden
NB 0098	DNV GL SE	Germany
NB 2388	DNV GL- Business Assurance	Netherlands
NB 2460	DNV Nemko Presafe AS	Norway
NB 2099	DNV Oil & Gas Denmark	Denmark

Principles

- ✓Products *placed on the Community market* shall comply with all applicable legislation (product directives).
- ✓The manufacturer, having detailed knowledge of the design and production process, is best placed to carry out the complete conformity assessment procedure. *Conformity assessment should therefore remain the obligation of the manufacturer alone*.
- ✓Standards and presumption of conformity
- ✓EC directives define the "essential requirements", e.g., protection of health and safety, that goods must meet when they are placed on the market.
- ✓The European standards bodies have the task of drawing up the corresponding technical specifications meeting the essential requirements of the directives, compliance with which will provide a *presumption of conformity* with the essential requirements. Such specifications are referred to as "*harmonised standards".*

Principles, contd.

✓Liability for defective products (producer liability)

The advantages of the Single market in facilitating the availability in all member States of a wide range of products from all corners of the Union must be made available within a framework which assures citizens of *the safety of those products and the possibility to claim compensation in case of damages caused by defective products*. Since 1985, the EU Internal Market counts on a framework of product liability set up to satisfy both concerns.

✓Directive 1985/374 and Directive 1999/34 apply to any product marketed in the European Economic area and is of direct concern for both citizens and producers in that area. By striking a fair balance of risk among citizens and producers, this legislation aims to converge consumers' interests with Single Market policies (namely free exchange of goods and elimination of competition distortions).

Declaration of Conformity

Content requirements given in the individual directive Annex IV of 2014/68/EU

Common template is here

Declaration of Conformity

1. Unique identification:
2. Name and address of (authorised representative of the) manufacturer:
3. This declaration of conformity is issued under the sole responsibility of the manufacturer (or installer):
4. Object of the declaration *(identification of product allowing traceability)*:
5. The object of the declaration described above is in conformity with the essential requirements of directive(s) ……
6. References to the relevant harmonised standards used, or references to the specifications in relation to which conformity is declared:
7. The notified body … *(name, number)*… performed … *(description of intervention)*… and issued the certificate: ….
8. Additional information:

   ……..

   ……..

   Signed for and on behalf of: …….
   (place and date of issue)

9. *(name, function)* *(signature)*

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- CE Marking and Introduction to PED - Scope and Purpose
- Annex II – Design & Categorization
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CE Marking

CE

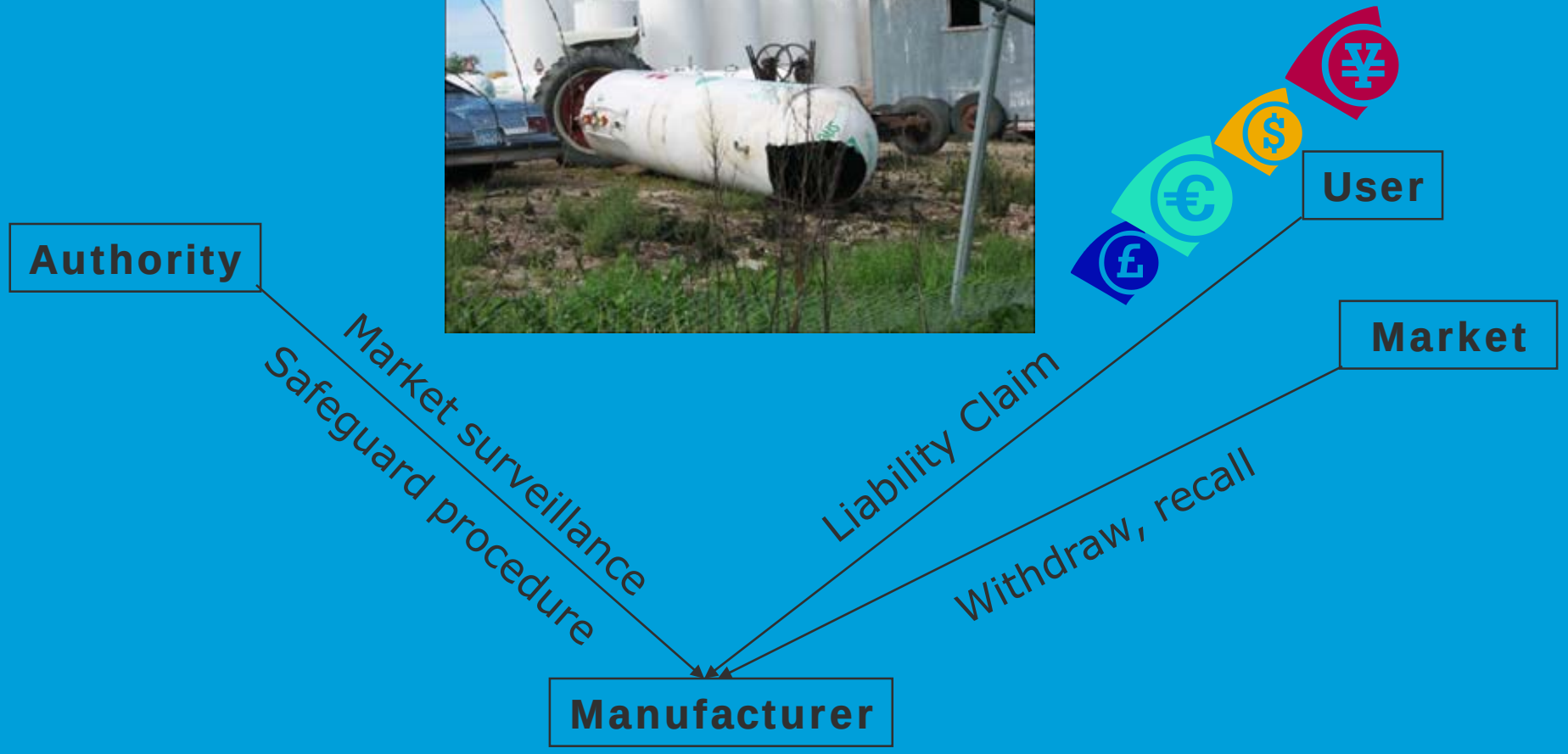
1234

The CE mark, denoting:
“Communauté Européenne” confirms that a product complies with all relevant EU Product Directives.

CE Marking declares the completion of conformity assessment and that the equipment or assembly complies with the provisions of the Directive and meets the **E**ssential **S**afety **R**equirements.

To be accompanied with the identification number for the Notified Body responsible for surveillance (assessment of the production), e.g. for DNV GL 1234

Non-Compliant Product



PED: Pressure Equipment Directive

http://ec.europa.eu/growth/sectors/pressure-gas/

The Blue Guide:

Blue Guide 2016
This Guide is intended to contribute to a better understanding of internal market legislation on industrial products in EU

Common Terminology used in PED and other EU Directives

- ✓ European Commission: Executive arm of the of the European Union, responsible for development and enforcement of EU Directives
- ✓ **N**otified **B**ody: a third party body, appointed by the national authority of an EEA member state, for performing conformity assessment for specific EU Product Directives
- ✓ Recognized Third Party Organisation : a third party appointed by the national authority of an EEA member state, specifically for approval of welding and NDE procedure and personnel approval.

‘*Harmonized*’ Product Standard: A technical standard, reviewed and approved by the European Commission, which, when used in full, satisfied all Essential Safety Requirements of relevant EU Product Directives.

‘*Harmonized*’ Support Standard: A technical standard, reviewed and approved by the European Commission, which satisfies specific requirements of EU Product Directives, such as issues related to welding, material specifications, NDE Operator qualifications. Example: EN ISO 15614 series (welding procedure qualification)

Each harmonized standard will contain an annex stating which essential safety requirements it satisfies. The advantage of using these standards is that no additional justification is required that essential safety requirements have been met and providing the standards have been correctly applied the design of the equipment can not be challenged.

Pressure equipment directive 2014/68/EU(97/23/EC)

- ✓ Began Transition period 29 November 1999
- ✓ Became Mandatory from 29 May 2002
- ✓ Applies to Pressure Vessels, Piping, and Pressure and Safety Accessories
- ✓ New PED 2014/68/EU became mandatory from 19 July 2016

Pressure equipment directive 2014/68/EU

- ✓Where to find information
- ✓Pressure Equipment Sector homepage in European Commission
- ✓http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:31997L0023
- ✓215 Guidelines as a help in interpreting the 2014/68/EU(PED 97/23/EC)
- ✓In order to ensure a coherent application of the Pressure Equipment Directive 2014/68/EU(97/23/EC PED), Guidelines are developed and agreed by the Commission's Working Group "Pressure" (WGP). This working group, created as a result of Article 17 of the PED, which requests the Member States to cooperate in order to assist the functioning of this Directive, is composed of representatives of Member States, European federations, the Notified Bodies Forum and CEN and chaired by a representative of the Commission services
- ✓The PED Guidelines are not a legally binding interpretation of the Directive. The legally binding text remains that of the Directive. However, the PED Guidelines represent a reference for ensuring consistent application of the Directive. They represent, unless indicated differently in the respective guideline text, the unanimous opinion of the Member States.

Scope and Purpose

The PEDirective applies to:

- ✓ The design, manufacture and conformity assessment of pressure equipment and assemblies with a maximum allowable pressure (PS) greater than 0,5 bar
- ✓ Every type of pressure equipment is covered if not specifically listed as excluded in Article 1, Sec. 3 (items 3.1 to 3.21)
- ✓ Also (as for all other directives) does not cover “in-service”

Scope and Purpose

The PEDirective applies to:

- ✓ Pressure vessels

- ✓ Piping

- ✓ Pressure accessories

- ✓ Safety accessories

Pressure Equipment Directive 2014/68/EU – PED (cont´d)

SCOPE/PURPOSE OF THE DIRECTIVE

'Assemblies` means several pieces of pressure equipment assembled by a manufacturer to constitute an integrated and functional whole.

Scope and Purpose

Case study I

- ✓ *safety valve ➔ safety accessory*
- ✓ *cover, collar, gasket, flange, bolt ➔ components of a pressure equipment*
- ✓ *sight glass with its frames ➔ components of a pressure equipment*
- ✓ *flanges, fittings, plain pipe lengths ➔ components of a pressure equipment*
- ✓ *Thermo wells ➔ components of a pressure equipment*
- ✓ *valves, meters ➔ pressure accessory (Guideline 2/31)*

Pressure Equipment Directive 2014/68/EU– PED (cont´d)

SCOPE/PURPOSE OF THE DIRECTIVE

Flanges, fittings, plain pipe lengths, heads and other equipment that don't have an identifiable pressure housing (such as Thermowells) are considered as "components" of pressure equipment and not equipment themselves, so they can not be CE marked for compliance with PED. (Guideline 4/9, 7/19)

Scope and Purpose

Case study I (Con't)

- ✓ *safety pressure relief systems ➔ safety accessory*
- ✓ *shutdown and lockout, such as pressure switches or temperature switches or fluid level switches ➔ safety accessory*
- ✓ *flow meter ➔ components of a pressure equipment*
- ✓ *connecting together piping components (flanges, pipes, fittings, reducers for example) ➔ item of piping (guideline 2/31)*
- ✓ *Pipe, tubing, hoses ➔ components of a pressure equipment (guideline 1/9)*

The PED hierarchy

A Pressure vessel is built up by:

Welding

Material

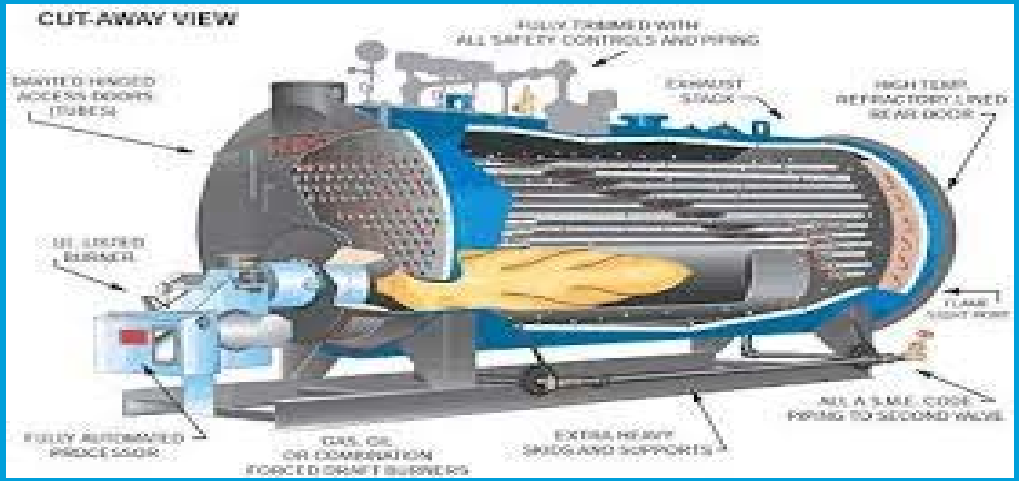
Components

- The material must be purchased with technical and formal requirements to PED ESR – No CE-marking
- The components are refined of correct material (forged, machined, welded) mostly according to a component standard – No CE-marking
- The pressure vessel manufacturer must reassure that this is covered in order to CE-mark – CE-marking
- There are strict requirements on welding (permanent joining) for the joining of all pressure bearing parts

Scope and Purpose

Case study II

pressure cookers, portable extinguishers, breathing apparatus, skid mounted systems, autoclaves; air conditioner, compressed air supply in a factory, refrigerating system, shell boilers, water tube boilers, distillation, evaporation or filtering units in process plants, oil heating furnaces ➔ assemblies (guideline 3/8)



Scope and Purpose

Notable Exclusions: (from Article 1, Section 3)

- ✓ Pressure Equipment on Ships or Mobile Offshore Units (*except FPSOs when considered "fixed"*)
- ✓ Well-control equipment, blow-out preventers, Christmas Tree, piping manifolds and related equipment
- ✓ Transmission pipelines to or from an installation (onshore or offshore), but includes compression or pressure reduction stations

Note) FPSO, FPP

A mobile offshore unit is an offshore unit that is not intended to be placed permanently or long term on the field, but is designed to be moved from location to location whether or not it has a means of propulsion or of lowering legs to the seafloor (e.g. a unit used solely for exploration).

For example, floating units intended for production, such as FPSO´s (Floating Production, Storage and Offloading installations usually based on tanker designs) and FPP´s (Floating Production Platforms based on semi-submersible vessels), are not considered to be mobile.

Note) Items of pressure equipment specifically intended for mobile offshore units are excluded from the PED.

However, items of pressure equipment intended to be installed on both FPSO´s/FPP´s and mobile offshore units are not excluded from the PED. (guideline 1/27)

PED - exclusions

Nuclear use items

pressure equipment consisting of a flexible casing

ADR

IMDG

ships, rockets,air craft and mobile off-shore units

radiators and pipes in warm water heating systems

RID

ICAO

Pipelines

Scope and Purpose

Notable Exclusions: (from Article 1, Section 3)

- ✓ Simple pressure vessel (SPVD), e.g portable air compressor
- ✓ Items specifically designed for nuclear use, failure of which may cause an emission of radioactivity;
- ✓ Pressure equipment consisting of a flexible casing, e.g. tyres, air cushions, balls used for play, inflatable craft, and other similar pressure equipment
- ✓ Transmission pipelines to or from an installation (onshore or offshore), but includes compression or pressure reduction stations
- ✓ Radiators and pipes in warm water heating systems
- ✓ Public water distribution networks

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Design & Categorization

- ✓ Identify applicable item of PED (PRESSURE VESSEL /PIPING/PRESSURE ACCESSORY/SAFETY ACCESSORY/ASSEMBLY)
- ✓ Design code selection
- ✓ Comply to design code +PED requirements
- ✓ PED is not a design code

Sample design codes:

- ✓ Pressure vessel—ASME Sec VIII Div 1,PD 5500/EN 13445
- ✓ Piping –ASME B31.3
- ✓ Pressure accessories—API 6A,API 6D,ASME B16.34
- ✓ Safety accessories-API 521
- ✓ Assemblies –Combination of above

Design & Categorization

- ✓ Design Review by NB depends on PED Module
- ✓ Design by Manufacturer/Design review by Notified Body for Cat II/III/IV
- ✓ Documents required to be submitted by Manufacturer TO Notified Body
  - ➢ Drawings/Design calculations
  - ➢ CE name plate drawing
  - ➢ Essential Safety requirements
  - ➢ Risk(Hazard)Analysis
  - ➢ PMA
  - ➢ Operating Instructions

DNV·GL

Design & Categorization

- ✓ To find out category and Module
- ✓ Is it a vessel/piping/pr. Accessory /safety accessory??
- ✓ What is the content? Is it gas/Liquid?
- ✓ What is the pressure(MAWP) and Volume if vessel?
- ✓ What is the Pressure and Diameter if piping?

Design & Categorization

Pressure equipment shall be classified by category in accordance with Annex II, according to ascending level of hazard

Most vessels and piping are classified on the basis of their maximum allowable pressure (PS), and their volume V or their nominal size (DN), as appropriate, and the group of fluids for which they are intended,

Nominal Size (DN) means a numerical designation of size which is common to all components in a piping system other than components indicated by outside diameters or by thread size.

Note) Consideration Volume and DN
In some cases both volume and DN are considered appropriate. In such cases, the pressure accessory must be classified in the highest category. ➔ Refer to the Case study III
In the case of valves, DN is normally the more appropriate.(guideline 2/17)

Design & Categorization

Classification of fluid group

Group 1

- ✓ Explosive
- ✓ Extremely flammable
- ✓ Highly flammable
- ✓ Flammable
- ✓ Very toxic
- ✓ Toxic
- ✓ Oxidizing
- ✓ (*guideline 2/7*)

Group 2

All other fluids, including steam

Commission Directive 2001/59/EC adapting to Council Directive 67/548/EEC relating to the classification, packaging and labelling of dangerous substances

http://eurlex.europa.eu/smartapi/cgi/sga_doc?smartapi!celexapi!prod!CELEXnumdoc&lg=en&numdoc=32001L0059&model=guichett

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Design & Categorization

Product Classifications and relevant tables

	VESSELS				STEAM GENERATORS	PIPING			
State of Contents	Gas or liquid whose vapour pressure> 0,5 at TS		Liquid			Gas or liquid whose vapour pressure> 0,5 at TS		Liquid	
Fluid Group	Dang	Oth.	Dang	Oth.		Dang	Oth.	Dang	Oth.
Refer to table	Table 1	Table 2	Table 3	Table 4	Table 5	Table 6	Table 7	Table 8	Table 9

Design & Categorization

Article 3.3 products

Pressure equipment and/or assemblies below or equal to the limits in sections 1.1, 1.2 and 1.3 and section 2 respectively must be designed and manufactured in accordance with the sound engineering practice of a Member State in order to ensure safe use. Pressure equipment and/or assemblies must be accompanied by adequate instructions for use and must bear markings to permit identification of the manufacturer or of his authorized representative established within the Community. Such equipment and/or assemblies must not bear the CE marking referred to in Article 15.

Design & Categorization

9 Tables:

Table 1 to 4 for pressure vessels
- Based on Volume (L) and Pressure (bar)

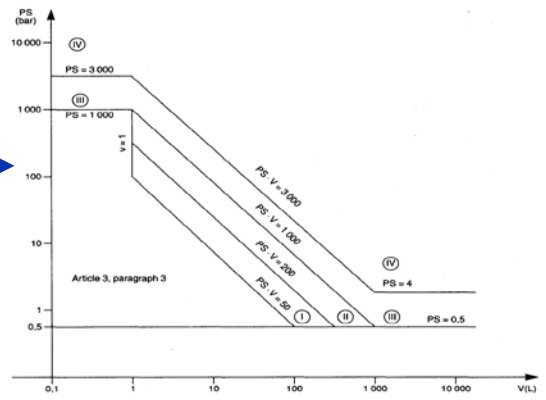


Table 5 for heated equipment with risk for overheating

Table 6 to 9 for piping mirrors table 1 to 4
- Based on dimension (DN) and Pressure (bar)

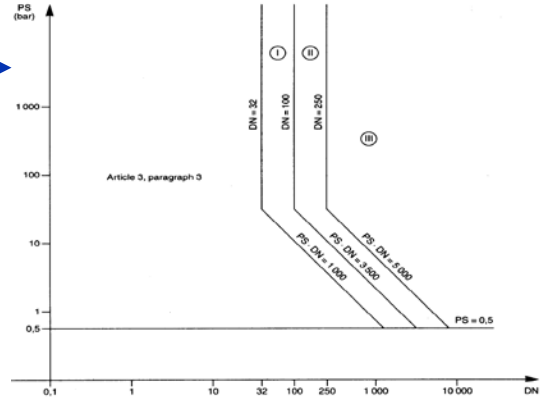
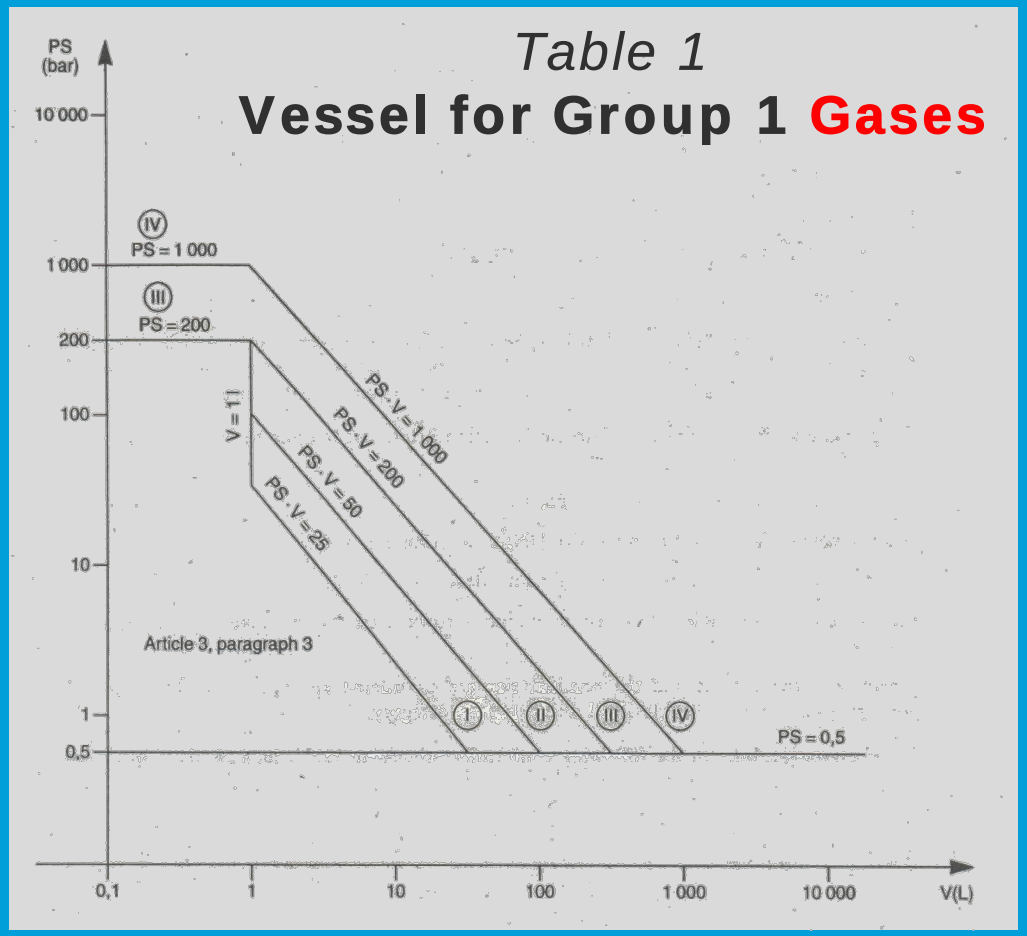


Table 7

The tables show Category I, II, III, IV for table 1 to 5 (vessels) and Category I, II, III for table 6 to 9(piping)

All tables show a low risk field called Article 3.3 or SEP

Design & Categorization



or **liquid** whose vapour pressure at the maximum allowable temperature is greater than 0,5 bar above normal atmospheric pressure,
Instable gas and falling within categories I or II on the basis of Table 1, must be classified in category III

Typical examples of unstable gases: acetylene (UN 1001), methyl acetylene (UN 1060), vinylf luoride (UN 1860).: Refer to Directive 67/548/EEC (guideline 2/21)

Design & Categorization

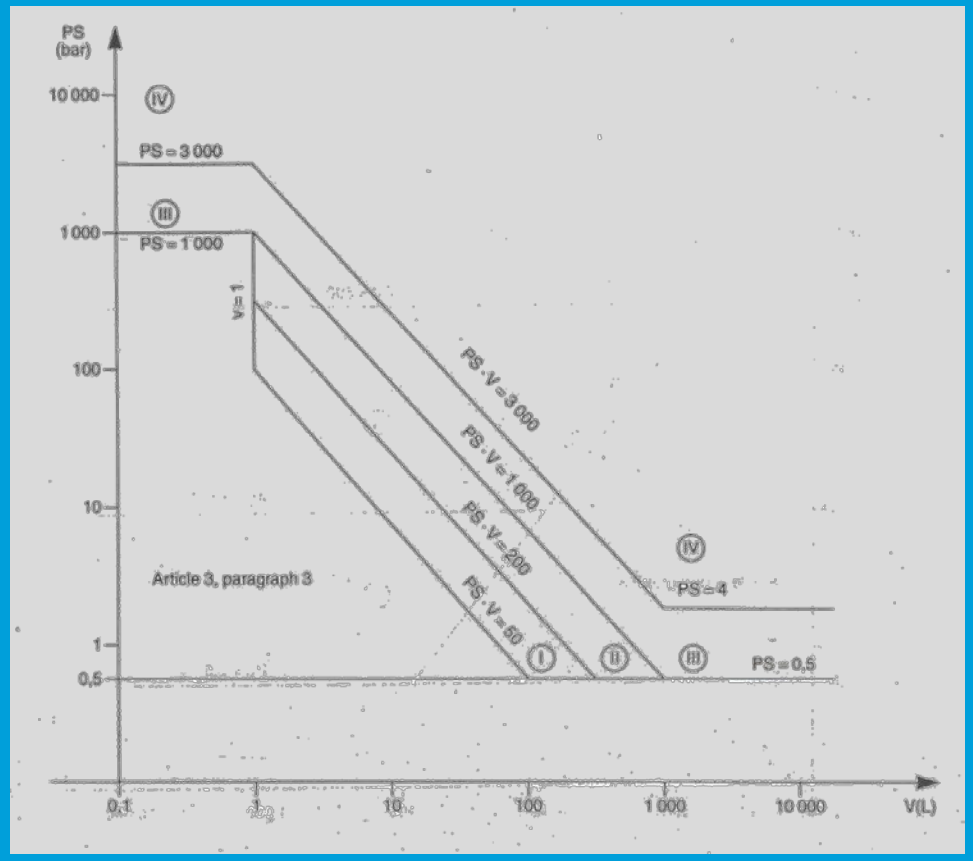


Table 2

Vessels for Group 2 Gases

or **liquid** whose vapour pressure at the maximum allowable temperature is greater than 0,5 bar above normal atmospheric pressure

Portable extinguishers and bottles for breathing equipment must be classified at least in category III

Design & Categorization

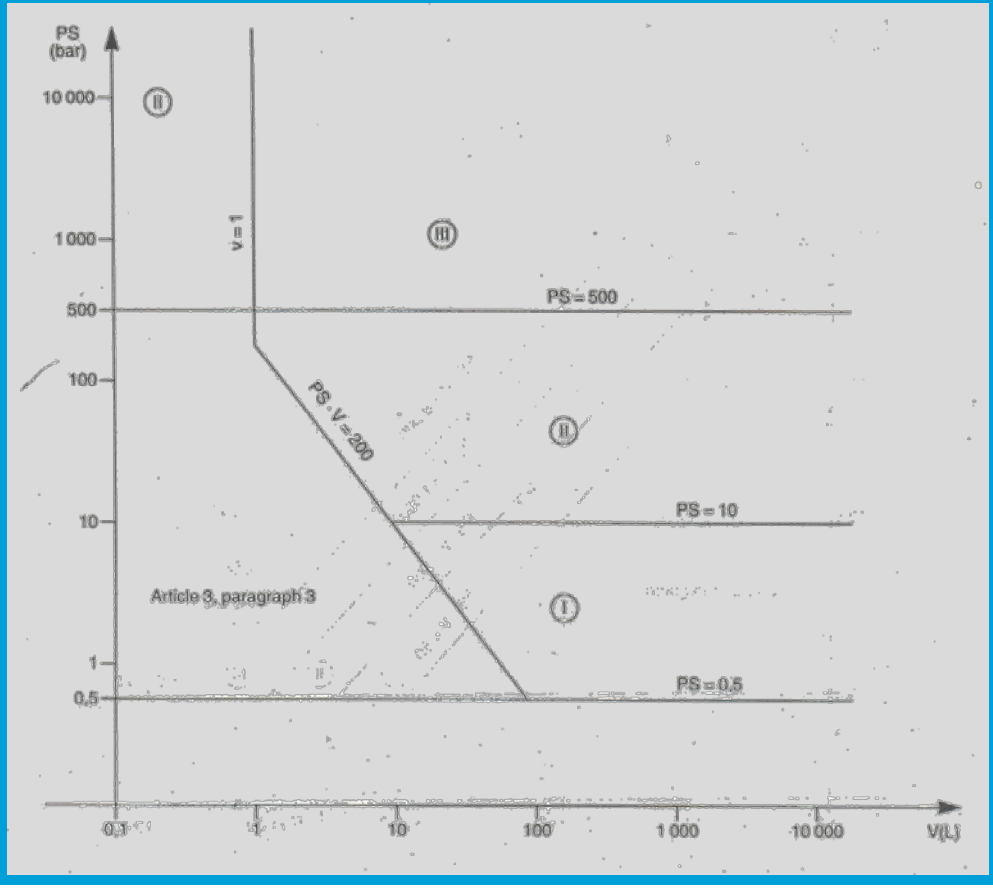


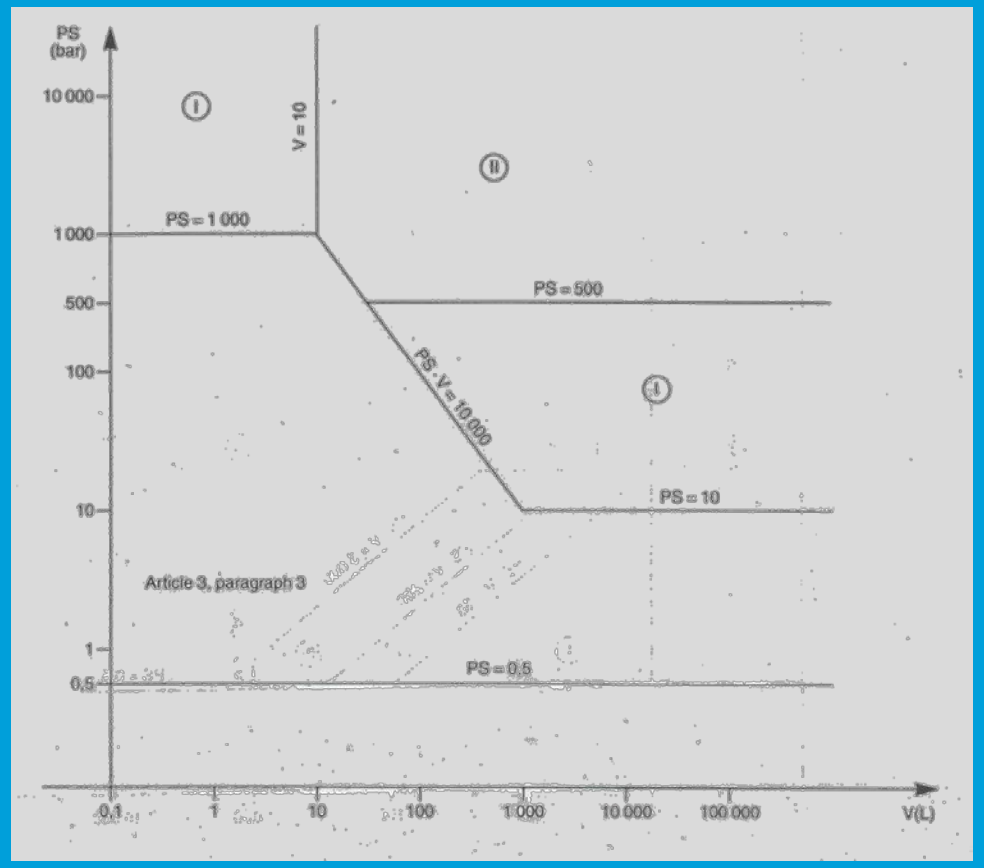
Table 3

Vessels for Group 1 Fluids

Design & Categorization

Table 4

Vessels for Group 2 Fluids



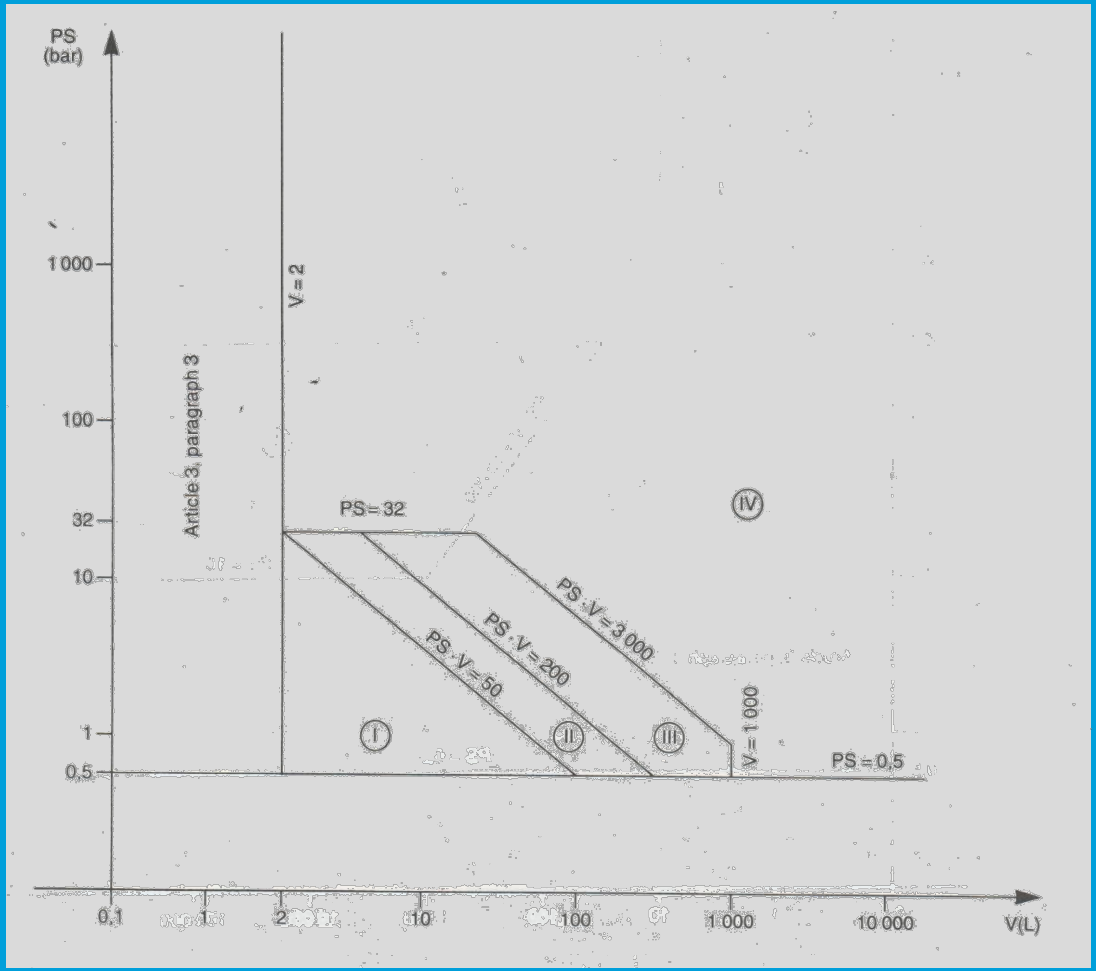
Assemblies intended for generating warm water at temperatures not greater than 110°C, which are manually fed with solid fuels and have a product or pressure and volume greater the 50 bar litres must be subjected to either Module B1 (for ESR's 2.10, 2.11, 3.4, 5 (a) (b), or Module H

Typical examples of unstable gases: acetylene (UN 1001), methyl acetylene (UN 1060), vinylfluoride (UN 1860).: Refer to Directive 67/548/EEC (guideline 2/21)

Design & Categorization

Table 5

Steam Generators

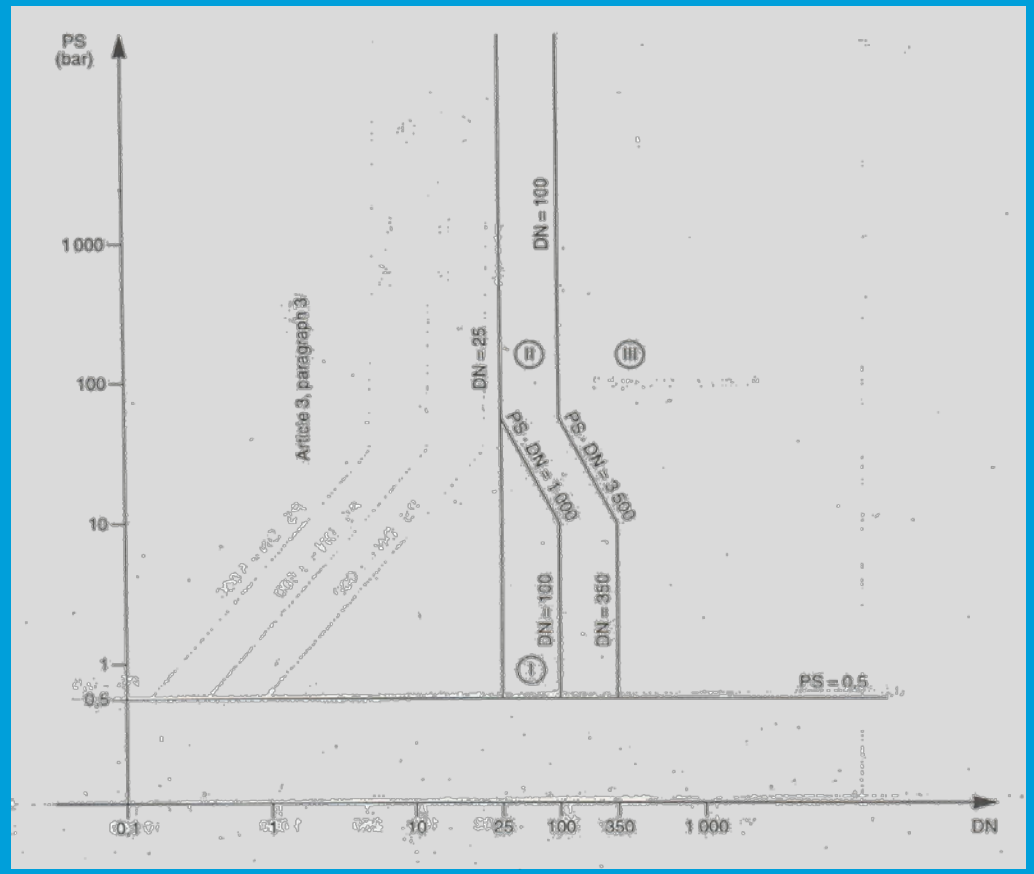


The design of pressure cookers must be subject to a conformity assessment procedure equivalent to at least one of the category III modules

Design & Categorization

Table 6

Piping for Group 1 Gases



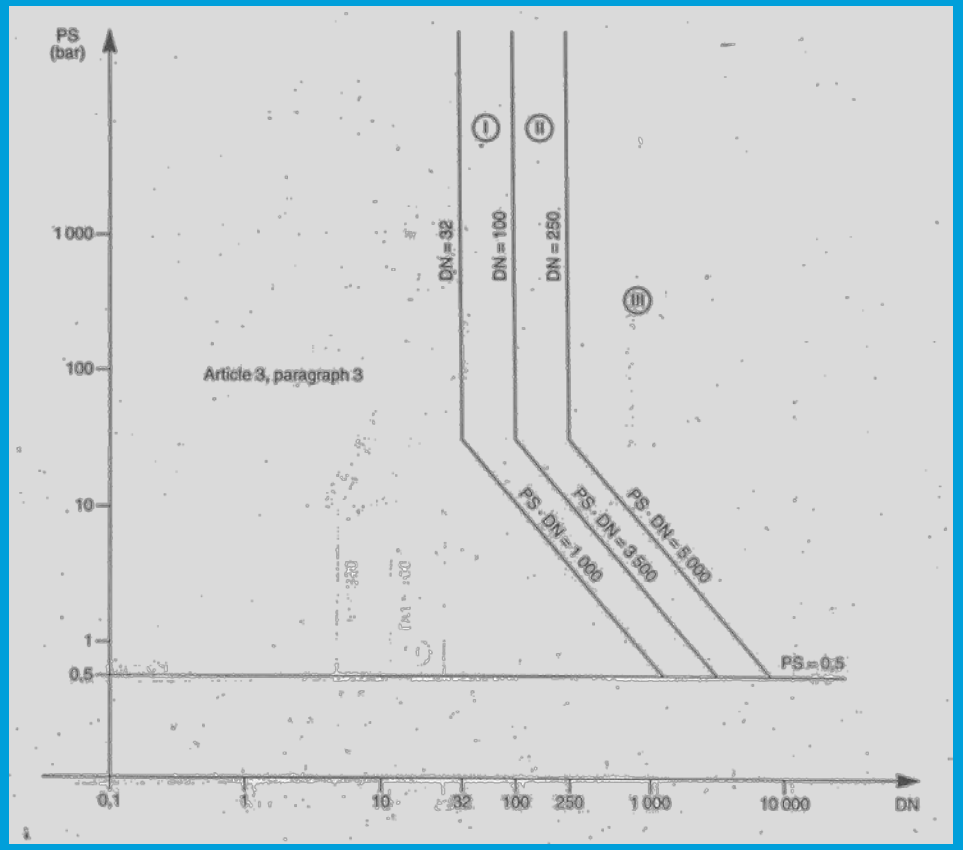
or **liquid** whose vapour pressure at the maximum allowable temperature is greater than 0,5 bar above normal atmospheric pressure

Instable gas and falling within categories I and II must be classified in category III

Design & Categorization

Table 7

Piping for Group 2 Gases



or **liquid** whose vapour pressure at the maximum allowable temperature is greater than 0,5 bar above normal atmospheric pressure

All piping containing fluids at a temperature greater than 350°C and falling within category II must be classified in category III

Design & Categorization

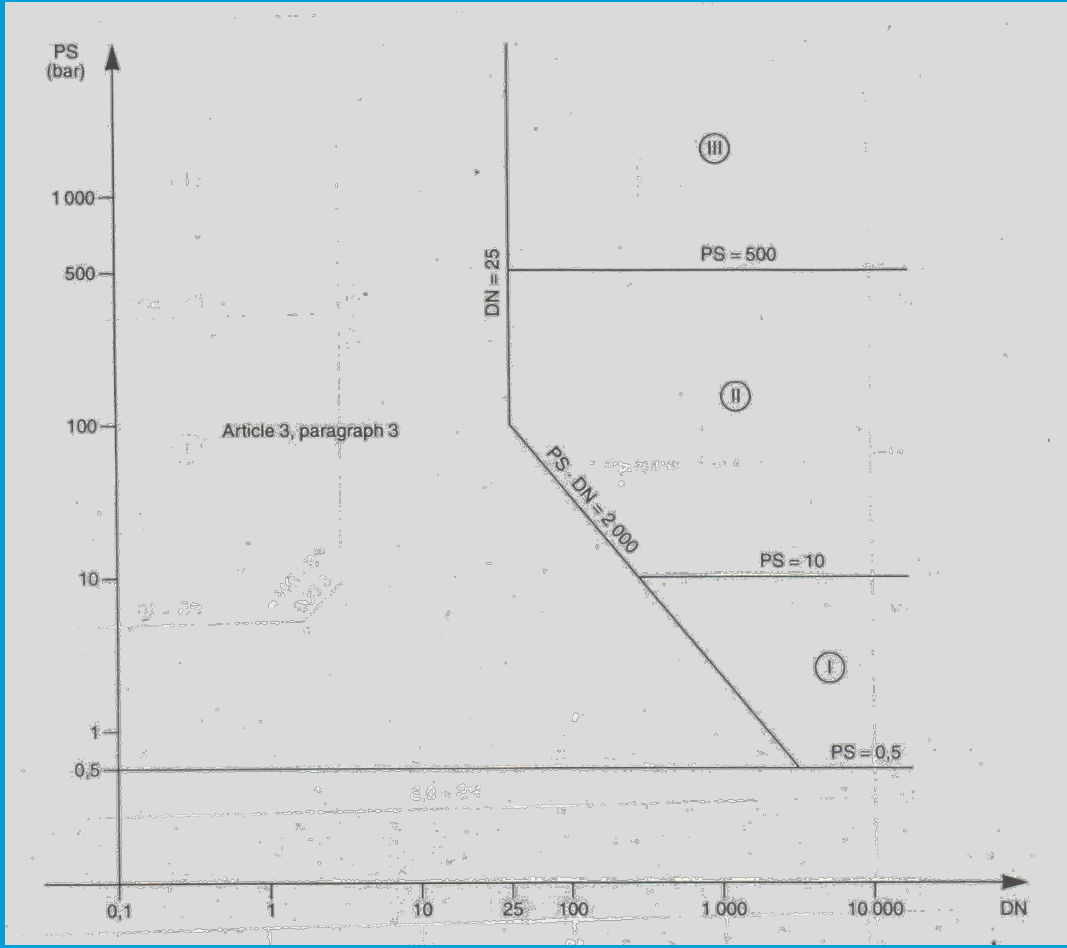


Table 8

Piping for Group 1 Liquids

Design & Categorization

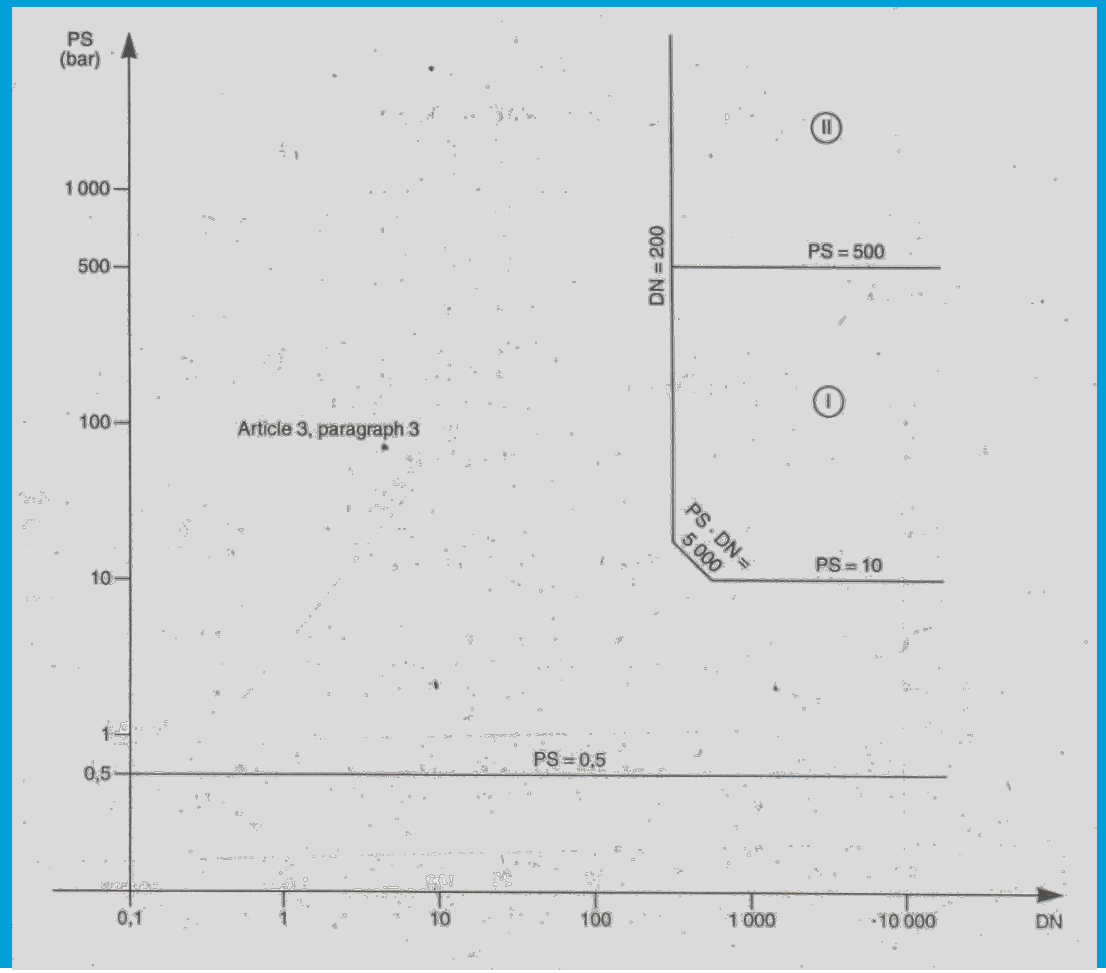
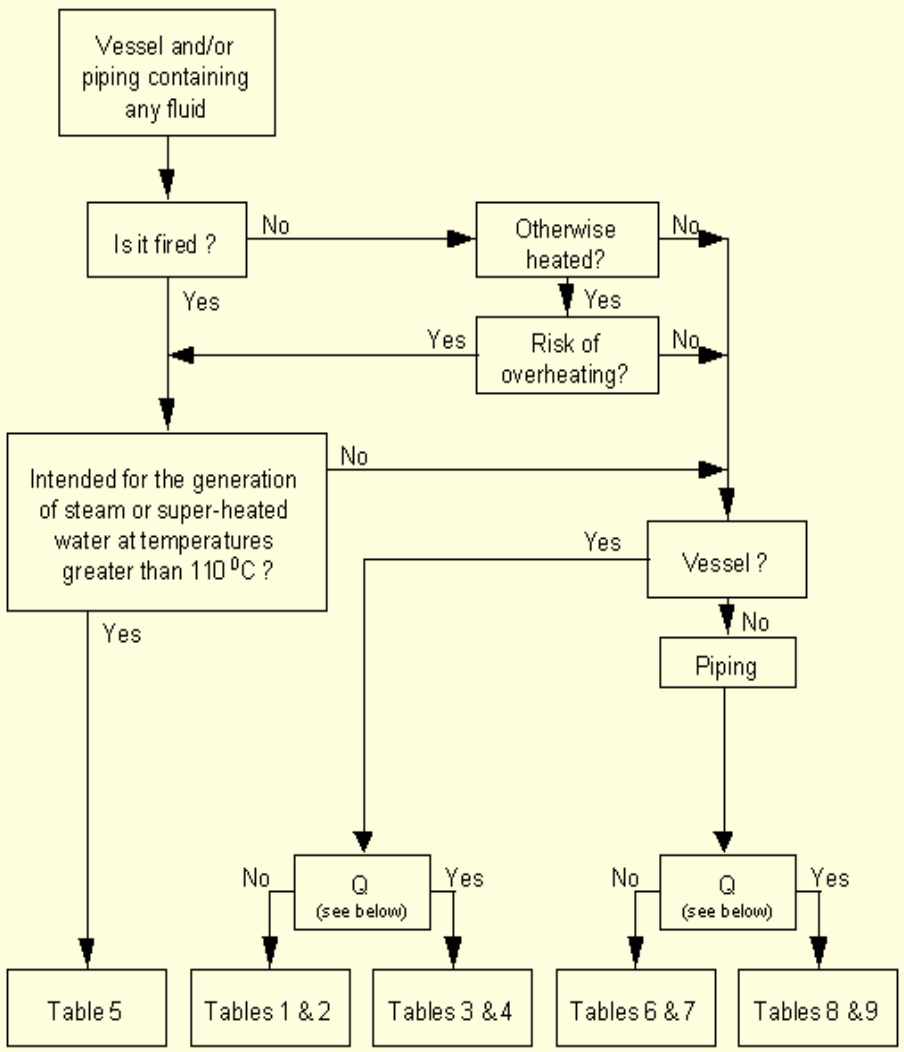


Table 9

Piping for Group 2 Liquids

Design & Categorization



Q. Does the vessel or piping contain liquid whose vapour pressure at the maximum allowable temperature is not more than 0,5 bar above normal atmospheric pressure?

Design & Categorization

Case Study III

Determine the equipment category within PED (annex II) for the following equipment:

Equipment:	**Well production test separator**
Pressure:	**350 PSI (24.13 bar)**
Volume:	**200 Gallon (757 litre)**
Contents:	**Group 1 fluid**

Design & Categorization

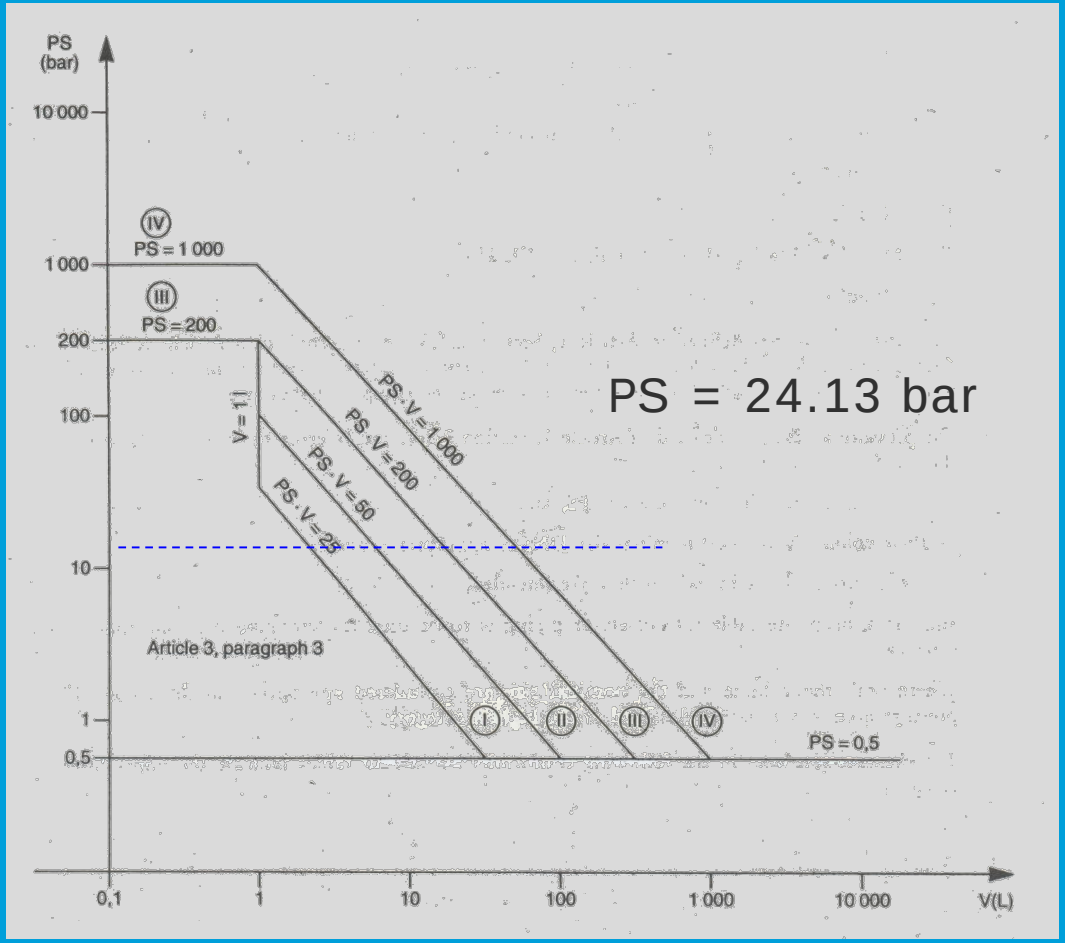
Case Study III

Evaluation Information:

Equipment Type	= **Vessel**
Fluid state:	= **Liquid & Gas**
Fluid group:	= **1**
Annex II table	= **1**
PS x V	= **18,266 (24.13 x 757)**

Design & Categorization

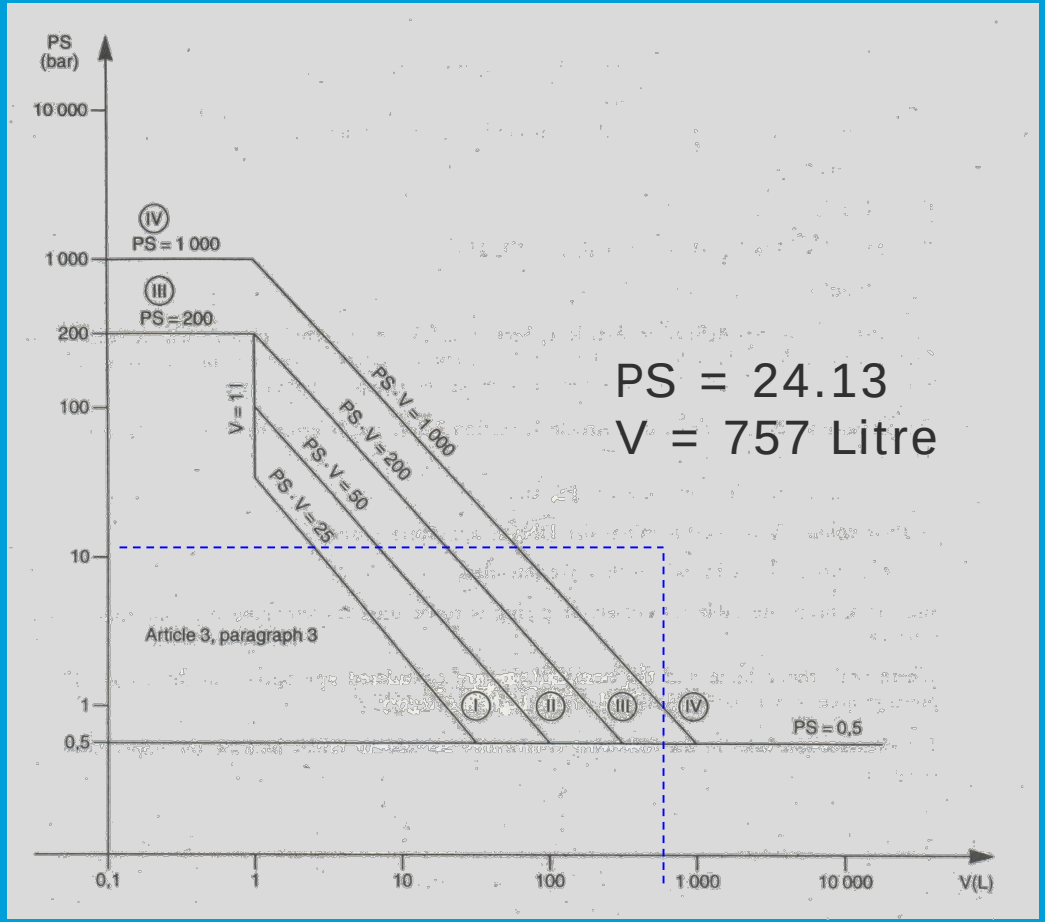
Case Study III



PS = 24.13 bar

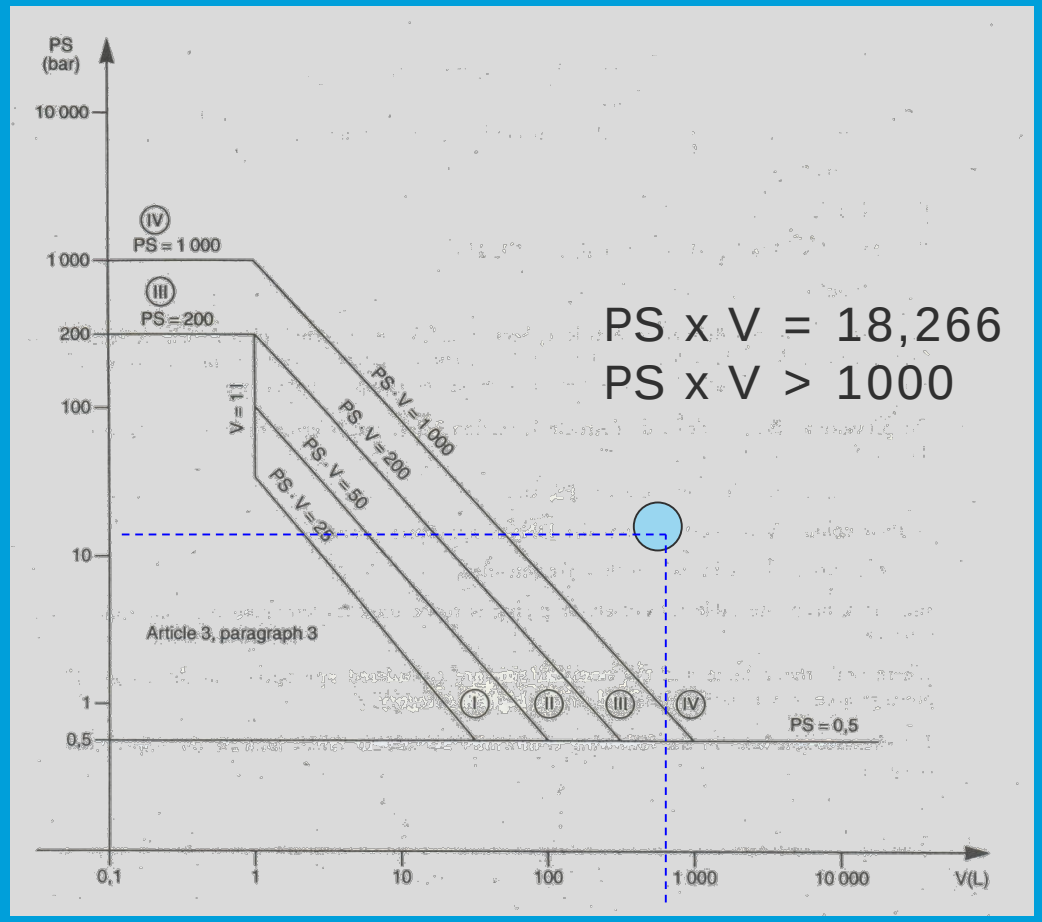
Design & Categorization

Case Study III



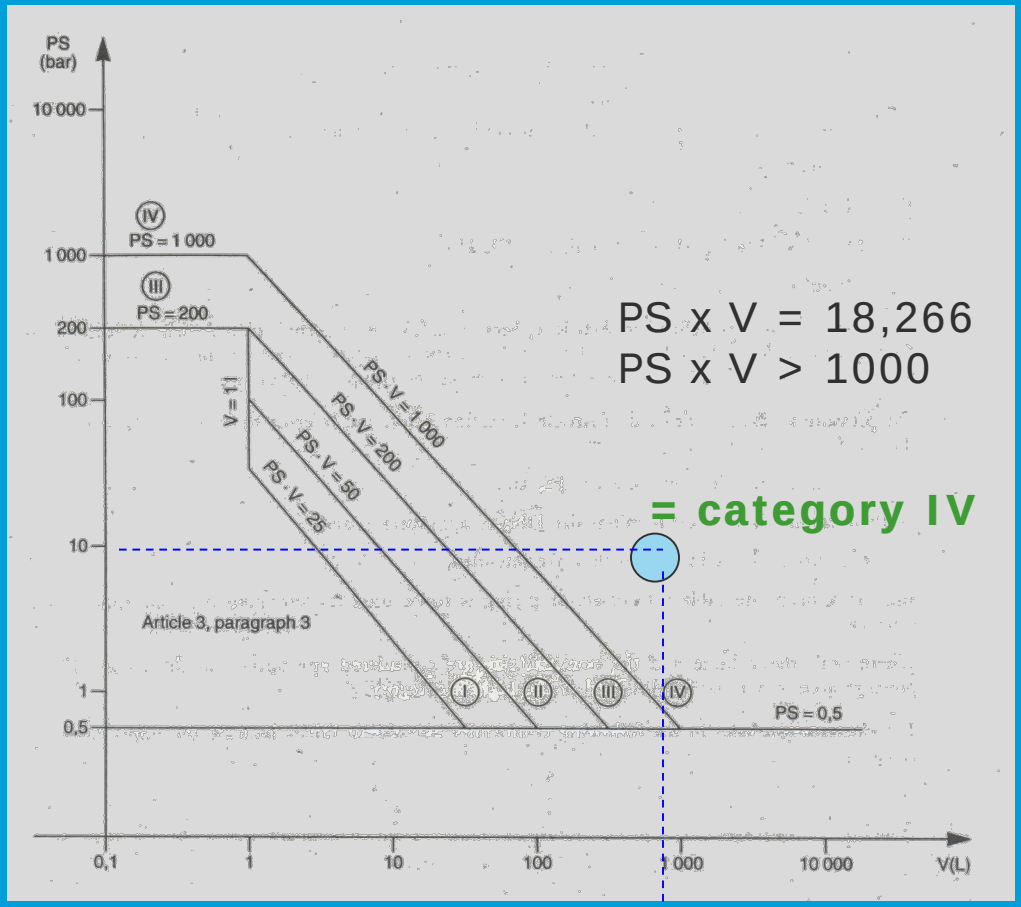
Design & Categorization

Case Study III



Design & Categorization

Case Study III



Conformity Assessment Modules as per 97/23/EC (Old)

Type Pressure Equipment			Category	Module for Conformity Assessment	Conformity Assessment Process per Annex III of the PED	
E.g. Pressure Vessels, Pressure Accessories, Pressure Assemblies	Piping		I	A	Internal Process Control	
			II	A1	Internal Process Control with Monitoring by Notified Body	
				D1	Production Quality Assurance	
				E1	Product Quality Assurance	
		Portable Extinguishers and Breathing Equipment	III	B1+D	EC Design Examination +	Production Quality Assurance
				B1+F		Product Verification
				B+E	EC Type Examination +	Production Quality Assurance
				B+C1		Conformity to Type
				H	Comprehensive Quality Assurance	
	Safety Accessories		IV	B+D	EC Type Examination +	Production Quality Assurance
				B+F		Product Verification
				G	EG Individual Testing	
				H1	Full Quality Assurance with Design Examination and Special Surveillance of the Final Product examination	

DNV·GL

Conformity Assessment Modules as per 2014/68/EU

1. The references in the tables to categories of modules are the following:

I	=	Module A
II	=	Modules A2, D1, E1
III	=	Modules B (design type) + D, B (design type) + F, B (production type) + E, B (production type) + C2, H
IV	=	Modules B (production type) + D, B (production type) + F, G, H1

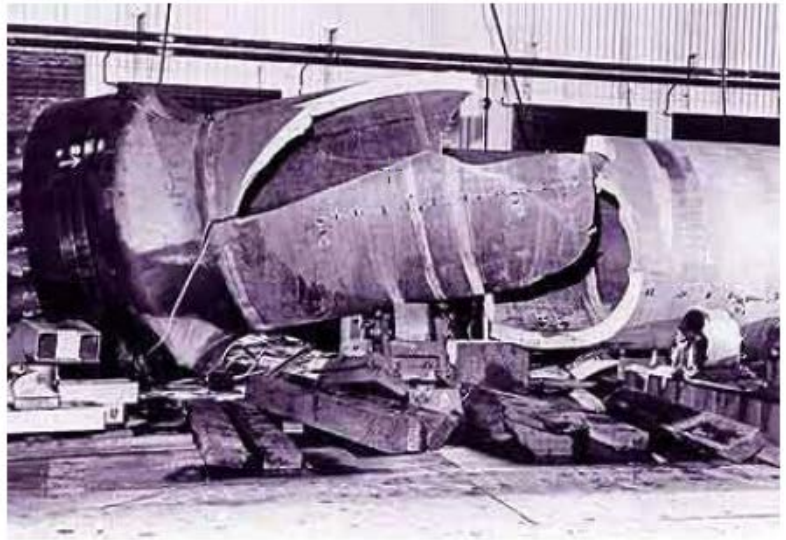
AGENDA

- Introduction to principles behind EU Product Directives (New Approach) / CE Marking
- Introduction to PED - Scope and Purpose
- Annex II – Design & Categorization
- Materials
- Permanent Joining (Welding) and NDE
- Final Assessment
- Annex I – ESRs
- Annex III - Conformity Assessment Modules
- PED in Practice
- Questions and Answers

Materials

Manufacturing Requirements

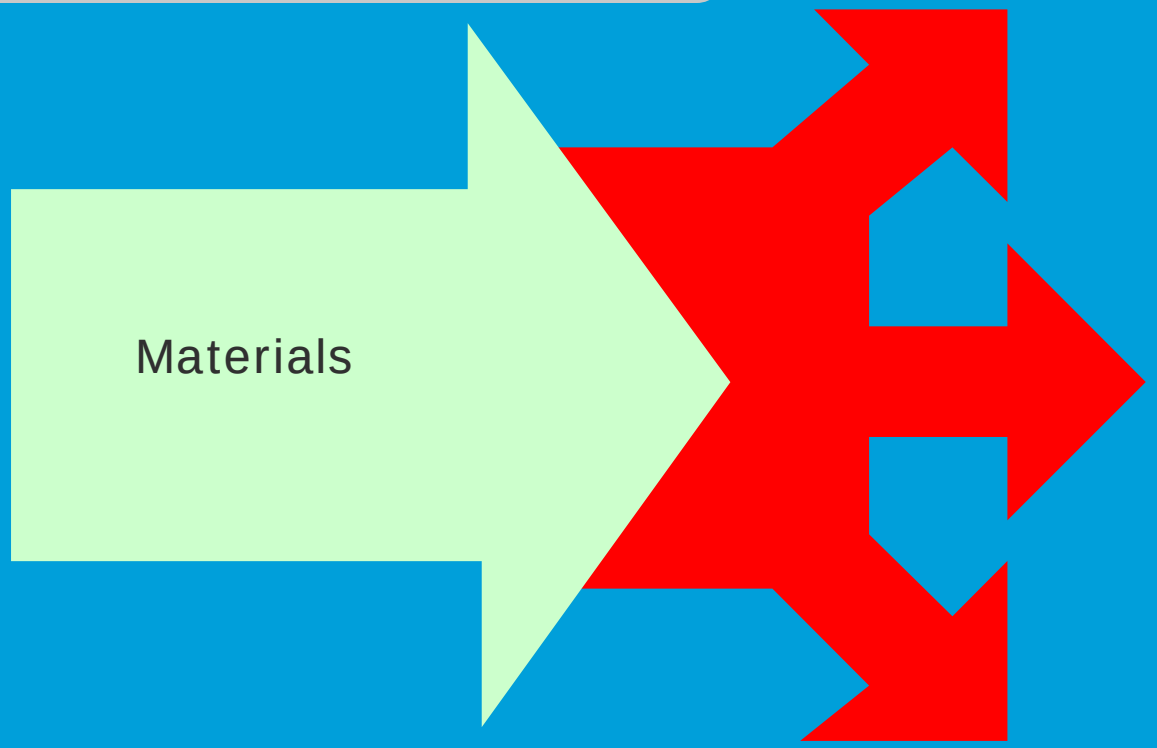
Designer must select materials that are suitable for the expected application and intended lifetime



Brittle fracture in crmov steel pressure vessel probably caused through poor toughness, high residual stresses and hydrogen cracking

Materials

Provide list of materials that demonstrates source:



Harmonized Standard

European Approval of Material

Particular Material Appraisal

Materials

Harmonized Standards

EN 10028 Pt 1 EN 10028 Pt 2 EN 10028 Pt 3 EN 10028 Pt 4 EN 10028 Pt 5 EN 10028 Pt 6 EN 10028 Pt 7	Steel Plates
EN 10269	Fasteners
EN 12420 EN 12451 EN 12452	Copper
EN 10213 Pt 1 EN 10213 Pt 2 EN 10213 Pt 3 EN 10213 Pt 4	Steel castings
EN 10222 Pt 1 EN 10222 Pt 2 EN 10222 Pt 3 EN 10222 Pt 4 EN 10222 Pt 5	Steel forging
EN 12482	Aluminium
EN 10216 EN 10217	Seamless & welded tubes

Materials

EAM

European Approval of Materials *is a technical document defining the characteristics of materials intended for repeated use in the manufacture of pressure equipment the type of which is NOT already covered by a harmonised standard. Notified bodies issue these documents. The notified body shall determine and perform the appropriate inspections and tests to certify the conformity of the types of material with the corresponding requirements of the PED.*

Materials

Manufacturing Requirements

Particular Material Appraisal (PMA)is the process by which the manufacturer ensures that each proposed material (that is not in a harmonized standard or covered by EAM) is:
Conforms to the applicable Essential Safety Requirements, or;
Achieves an “equivalent level of safety” to ESR’s
For pressure equipment in Category III and IV, a Notified Body must also perform this appraisal.

NOTE: It is always the manufacturer’s responsibility to confirm the material is suitable for it’s intended use.

Materials

Main issues concerning PMA's:

Approval of ASME materials for use with PED:

- ✓ Impact toughness
- ✓ Ductility
- ✓ Forming- and welding
- ✓ Not affected by ageing

Materials

Manufacturing Requirements

Approval of ASME materials for use with PED:
Generally, DNV GL will accept ASME/ASTM materials for use, through *particular material appraisal* when:

- Materials supplied meet mechanical requirements of PED (where appropriate)
- ASME or suitable recognized design and fabrication code is used in it's entirety
- ASME welding procedures are used, often in conjunction with EN ISO 15614 series supplemental testing.

NOTE: some Notified Bodies will not accept ASME material properties or chemistry

Materials

PMA Example

Particular Material Appraisal ASTM B150, type UNS C63000	Date: 2008-01-29 Revision: 0 Page: 1 (1)

Pressure Equipment Manufacturer:			Applicable Design Code:
Gusto B.V., the Netherlands			ASME B31.3
Material specifications:	Grade:	Product form:	Heat treatment condition:
ASTM B150	UNS C63000	Stud bolts	HR50
Application:			Max. allowable pressure [bar]
Applications covered by Directive 97/23/EC			According to ASME B31.3
Min. allowable temperature [°C]:			Max. allowable temperature [°C]
According to ASME B31.3			According to ASME B31.3

Compliance with Essential Safety Requirements

Property	Requirement	Compliance
Appropriate properties (Annex 1 § 4.1 (a))	e.g. $R_{e/t}$, $R_{m/20}$, if required $R_{m/t}$	See "*Additional material req. 1*"
Sufficiently ductile (Annex 1 § 4.1 (a)) (Annex 1 § 7.5)	In steel, 14 %	Yes
Sufficiently tough (Annex 1 § 4.1(a), Annex 1 § 7.5)	e.g. in steel, 27 Joule at 20 °C or lowest operating temp.	Yes
Not significantly affected by ageing (Annex 1 § 4.1 (c))	At intended operating temperatures	Yes
Suitable for intended processing (Annex 1 § 4.1 (d))	e.g. welding, forming, etc	Yes
Material Marking (Annex 1 § 3.1.5)	Identifiable to Certificate	Yes
Type of Certificate (Annex 1 § 4.3)	Certified QA-system	Yes
Delivery conditions (Annex 1 § 4.3)	Affirmation of compliance with order	Yes

Additional material requirements:

1. *ASTM B150 does not specify elevated temperature properties for this steel. However, the design conditions specified in ASME B31.3 are found acceptable.*

Materials

Manufacturing Requirements

Materials must have appropriate properties for all reasonably foreseeable operating and test conditions, in particular concerning ductility and toughness. Where appropriate, material characteristics should meet: (accordance with 2014/68/EU(97/23/EC)Annex I clause 4 and clause 7.5)

- ✓ Elongation after rupture exceeds **14%**
- ✓ Charpy impact energy is not less that **27J**, at temperature not higher than lowest scheduled operating temperature, *but not higher than* **20°C**
- ✓ Sufficiently chemically resistant to the fluid contained in the equipment
- ✓ Not sufficiently affected by ageing
- ✓ Suitable for the intended processing procedures
- ✓ Selected in order to avoid significant undesirable effects when put together with other materials

Materials

Manufacturing Requirements – example comparison

		Steel designations		Outokumpu steel names	Typical chemical composition, %						National steel c
		EN	ASTM		C	N	Cr	Ni	Mo	Others	JIS/Japan
WET CORROSION AND GENERAL SERVICE	Ferritic	1.4016	430	4016	0.04	–	16,5	–	–	–	SUS 430
	Mart.	1.4006	410	4006	0.15	0.04	12	–	–	–	SUS 410
		1.4005	416	4005	0.10	0.04	13	–	–	S	SUS 416
		1.4021	420	4021	0.20	–	13	–	–	–	SUS 420J1
		1.4028	420	4028	0.30	–	12.5	–	–	–	SUS 420J2
		1.4418	–	248 SV	0.03	–	16	5	1	–	–
	Duplex	1.4162	S32101	LDX 2101®	0.03	0.22	21.5	1.5	0.3	5Mn	–
		1.4362	S32304	SAF 2304®	0.02	0.10	23	4.8	0.3	–	–
		1.4462	S32205*	2205	0.02	0.17	22	5.7	3.1	–	SUS 329J3L
		1.4410	S32750	SAF 2507®	0.02	0.27	25	7	4	–	–
	Austenitic	1.4310	301	4310	0.10	–	17	7	–	–	SUS 301
		1.4318	301LN	4318	0.02	0.14	17.7	6.5	–	–	SUS 301L
		1.4372	201	4372	0.05	0.15	17	5	–	6.5Mn	SUS 201
		1.4301	304	4301	0.04	–	18.1	8.3	–	–	SUS 304
		1.4307	304L	4307	0.02	–	18.1	8.3	–	–	SUS 304L
		1.4311	304LN	4311	0.02	0.14	18.5	10.5	–	–	SUS 304LN
		1.4541	321	4541	0.04	–	17.3	9.1	–	Ti	SUS 321
		1.4550	347	4550	0.05	0.04	17.5	9.5	–	Nb	SUS 347
		1.4305	303	4305	0.05	–	17.3	8.2	–	S	SUS 303
		1.4303	305	4303	0.04	–	17.7	12.5	–	–	SUS 305J1
		1.4306	304L	4306	0.02	–	18.2	10.1	–	–	SUS 304L
		1.4567	S30430	4567	0.01	–	17.7	9.7	–	3Cu	SUS XM7
		1.4401	316	4401	0.04	–	17.2	10.2	2.1	–	SUS 316
		1.4404	316L	4404	0.02	–	17.2	10.1	2.1	–	SUS 316L
		1.4436	316	4436	0.04	–	16.9	10.7	2.6	–	SUS 316
		1.4432	316L	4432	0.02	–	16.9	10.7	2.6	–	SUS 316L
		1.4406	316LN	4406	0.02	0.14	17.2	10.3	2.1	–	SUS 316LN
		1.4429	S31653	4429	0.02	0.14	17.3	12.5	2.6	–	SUS 316LN
		1.4571	316Ti	4571	0.04	–	16.8	10.9	2.1	Ti	SUS 316Ti
		1.4435	316L	4435	0.02	–	17.3	12.6	2.6	–	SUS 316L

Materials

Manufacturing Requirements

Recognized types of material certificates for PED: (EN 10204:2004)

- ✓ **Type 2.1** 'Declaration of Compliance' – manufacturer declares the product supplied are in compliance with the requirements of the order.
- ✓ **Type 2.2** 'Test Report' - the manufacturer certifies that the products supplied are in compliance with the requirements of the order and in which he *supplies test results based on non-specific inspection and testing*.
- ✓ **Type 3.1** ' Inspection Certificate 3.1' (old *3.1.B*) - issued by the department independent of manufacturing, based on inspection and testing carried out on the products supplied. (*specific inspection*)
- ✓ **Type 3.2** 'Inspection Certificate 3.2' (old *3.1C*) - issued by the material manufacturer and validated by an authorized representative of the purchaser, in accordance with the specifications of the order, based on inspection and testing carried out on the products supplied.

Materials

Manufacturing Requirements

Material documentation - Main Pressure Equipment

- ✓ Equipment category I ➔ Material Certificate type 2.2 or 3.1
- ✓ Equipment category II, III, IV, when the material manufacturer's quality system is certified by a "competent body established in the community, and having under gone a special assessment for materials" ➔ Material Certificate 3.1
- ✓ Equipment category II, III, IV, when the manufacturer's quality system is not certified by a "competent body established within the community" ➔ Material Certificate 3.2 issued by the material manufacturer <u>and</u> DNV GL (or recognized competent third party by DNV GL)

Materials

Manufacturing Requirements

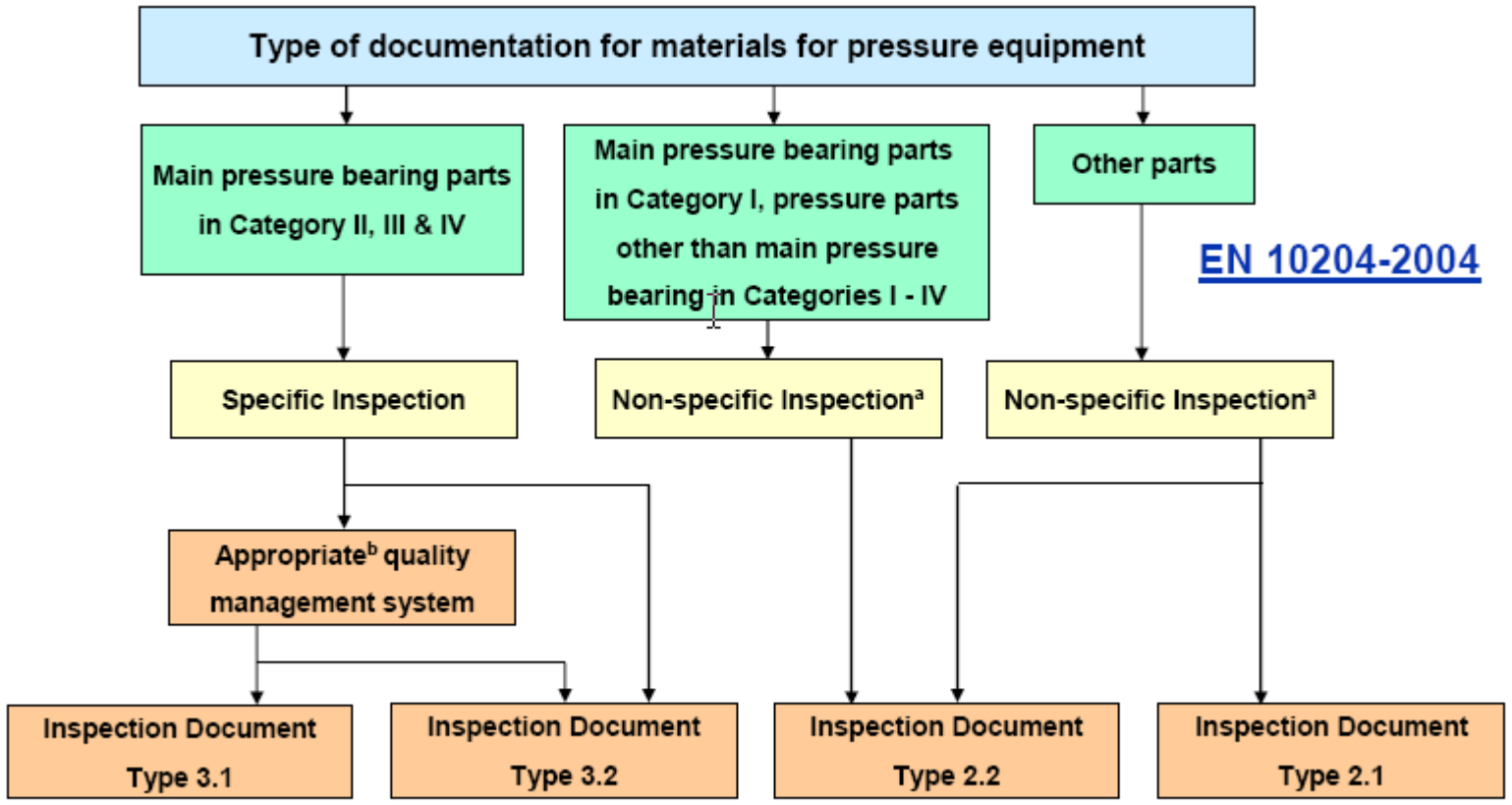
Pressure equipment parts may be divided into 3 groups (guideline 7/6, guideline 7/8) :

- ✓ **Main pressure-bearing parts**: Main pressure parts forming the pressure envelope, directly loaded by pressure (shells, domed ends, nozzle pipes, main body flanges, tube sheet of exchangers, tube bundles, clamp bolts, flange bolts etc.) ➔ Type 3.1 or 3.2
- ✓ **Pressure-bearing parts**: Parts indirectly loaded by pressure, such as stays, braces, doubling plates, flange bolts, etc. ➔ Type 2.2
- ✓ **Other parts**: Parts which loads are not originating from pressure, such as doubling plates for supports, lifting lugs, supporting structures, etc. ➔ Type 2.2

Note 1: ***Process of PMA for PE parts is same with Main Pressure Equipment***
NOTE 2: Certain parts may also be covered by other national legislation in the country of use, such as lifting lugs, support structures, etc. Fulfilment of those requirements is the responsibility of the supplier, and verification is not part of the Notified Bodies scope.

Materials

Material documentation - Main Pressure Components:



a Non-specific inspection may be replaced by specific inspection if specified in the material standard or the order

b Quality management system of the material manufacturer certified by a competent body established within the community and having undergone a specific assessment for materials

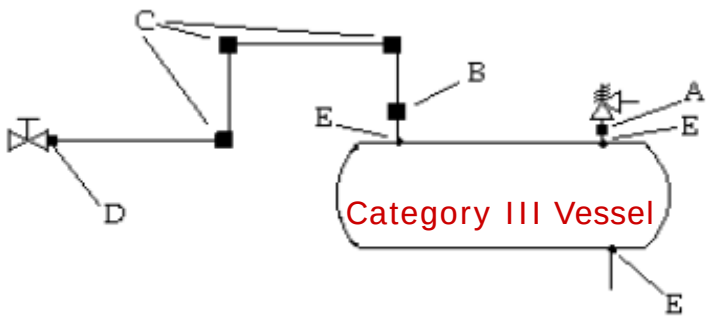
Materials

Case Study VI-1

No	Parts	Items	Required Material Certificate Type
1	Penetration Detail of Center Flange	Sleeve (for protecting pipe)	2.2
		Support structure	2.2
2	Flange Joint	Flange	3.1 or 3.2
		Bolt	3.1 or 3.2
		Gasket	3.1 or 3.2
3	Pressure Gage Assembly	Nipple	3.1 or 3.2
		on/off manual valve	3.1 or 3.2
		Gage	CE marking
		Boss	3.1 or 3.2
4	Pipe Support with U–Bolt	U–bolt	2.2
		Nut	2.2
5	Anchor with Support		2.2
6	Pad Eye on the pipe		2.2

Materials

Case Study VI-2



Joint reference	Description of the joints	Category of the joint
A	Connecting a category IV safety accessory to a category III vessel	III
B	Connecting a Category I piping to a Category III vessel (via a nozzle)	III
C [1)]	Integral joints of Category I piping	I
D	Connecting a Category I pressure accessory to a Category I piping	I
E [1)]	Connecting a nozzle to the shell of the category III vessel	III
[1)] *Integral joint of items of pressure equipment*		

Materials

What do DNV GL looks for in a Material Test Report/Inspection Certificate?

- ✓ Name & Address of Material Manufacturer
- ✓ Description of product
- ✓ Description of product specification, purchase conditions, etc. (Material specification, type , grade, heat treatment condition, etc. as relevant)
- ✓ *Type 2.2 Test Report*: Results of non-specific testing
- ✓ *Type 3.1 & 3.2 Inspection Certificate*: Results of batch specific testing
- ✓ Statement affirming compliance with purchase conditions or specifications
- ✓ Reference to certification of quality system (not mandated, but helpful)
- ✓ Signature and particulars of issuing party (co-signature of 3rd party for type 3.2 Inspection Certificate)

Materials

Case Study VII

Who is responsible for the PMA?

For PED Category I and II,

the PMA is under the responsibility of the equipment manufacturer

For PED Category III and IV,

the PMA is appraised by the nominated Notified Body

AGENDA

- Introduction to principles behind EU Product Directives (New Approach) / CE Marking
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Welding & NDE

Welding (joining) Procedures

- ✓ The properties of welded joints must meet the minimum properties for the base materials joined.
- ✓ Welding procedures must be supported with qualification tests
- ✓ **Welding procedures** and **personnel** for (pressure welds) on Category II, III and IV equipment must be approved by either a **N**otified **B**ody or a **R**ecognized **T**hird **P**arty **O**rganisation (Annex I 3.1.2)

NOTE: Third Party Organizations may appoint organisations according to Article 13 of the Directive, for specific tasks related to approval of welding procedures, personnel, and NDE operators. These appointments are listed with the Commission, in the same way as Notified Body appointment.

WPQ= Welder Performance Qualification
WPS= Welding Procedure Specification
PQR= Procedure Qualification Record

Welding & NDE

Welding Consumables (guideline 7/12)

Manufacturers of welding consumables shall provide inspection documents affirming compliance with the specification.

Based on section 4 of Annex I and guideline 7/5 manufacturers of welding consumables shall provide test report "2.2" as an inspection document in accordance with the standard EN 10204.

The traceability requirement of Annex I section 3.1.5 applies also for welding consumables. It can be maintained by procedural methods that cover receipt, identification, storage, transfer to production, temporary storage and use in production, availability of correct inspection documents at the final inspection (see also guideline 7/4).

NOTE :Welding consumables are treated as materials and must fulfil the relevant requirements of: 4.1, 4.2(a), 4.3

First paragraph, Certificate of conformance required

Welding consumables are defined by trade name, designation and relevant EN classification standard.

Inspection documents of welding consumables should give test results, for technical characteristics according to designation and classification standard, such as:

- ✓ Chemical composition of welding filler metal or all-weld metal as appropriate
- ✓ Tensile properties of all-weld metal: tensile and yield strength, elongation
- ✓ Impact properties of all-weld metal at temperature according to designation. Test results are based on non specific inspection and testing. They can be given for example as typical values based on quality control tests.

Welding & NDE

Welding (joining) Procedures

- ✓ Welding procedures and personnel qualifications are normally to be in accordance with Harmonized Standards, such as ISO 9606(EN 287)(Qualification test of welders)/ EN ISO 15614 (Specification and qualification of welding procedure), or equivalent standards. (guideline 6/8)

- ✓ When Notified Body approval is required, the Notified Body is responsible to determine the extent with which they must witness or conduct the appropriate tests. (guideline 6/1)

***Note**: EN ISO 15614-1 replaced EN 288-3 (Specification and approval of welding procedures for metallic materials)*

Welding & NDE

Welding (joining) Procedures

- DNV GL recognises ASME Section IX, usually when supplemental essential variables are applied, and combined with any relevant testing that would be required from a Harmonized Standard, such as EN 287/ EN ISO 15614 or EN 13445(unfired pressure vessels), as long as an ASME design and fabrication code is the basis of product design.

Note: NORSOK STANDARD M-601 (Welding and inspection of piping)

Welding & NDE

NDE Operator Qualification

- ✓ NDE Operators must be approved by a third-party organization recognized (RTPO) by a Member State for Category III, IV equipment.
- ✓ Typically recognized qualification schemes: PCN,EN 473, ISO 9712.
- ✓ ASNT Central Certification Program
- ✓ *The independent certifying bodies (ICB's) shall perform to the requirements in EN 45013. The certificates shall be reviewed and endorsed by a third party recognized by a member state.*
- ✓ Visual Examiners (VT) do not require formal qualification

NOTE: *Operators that do not possess certificates from above schemes must be Certified/Approved by an RTPO*

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Final Assessment

Final Assessment: Inspection and Testing

Equipment must undergo a final inspection and documentation review . Final inspection must be carried out internally and externally on every part of the equipment (at the appropriate stage of manufacturing)

Final Assessment

Final Assessment: Inspection and Testing

Testing:

Final assessment of equipment must include be pressure testing:

- ✓ Hydrostatic testing at $P_T = 1{,}43 P_S$
- ✓ If hydrostatic testing is harmful or impractical, other recognized tests may be carried out, but may require additional NDE to be carried out. (e.g.. pneumatic pressure test *guideline 8/16*)

*P*t is the test pressure measured at the highest point of the vessel in test position;

*P*s is the maximum allowable pressure of the vessel;

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Annex I - ESRs

General Requirements

- ✓ Eliminate or reduce hazards
- ✓ Apply protective measures for hazards not eliminated
- ✓ Inform users of residual hazards and required special measures necessary
- ✓ Design to prevent danger from ***known or clearly foreseeable*** misuse

The manufacturer is under an obligation to analyse the hazards, in order to identify those which apply to his equipment; he must design and construct it taking account of his analysis.

Annex I - ESRs

Risk(Hazard) Analysis

- ✓ Determination of the boundaries of the pressure equipment
- ✓ Identification of potential hazards (including misuse)
- ✓ Risk Evaluation

Notes:

Consideration of service condition must be made in each case of use, to select the most appropriate safety concept.

A consistent approach should be used by the manufacturer for risk analysis.

Annex I - ESRs

The **hazard analysis** should be a sequence of logical steps, which permit a systematic examination of hazards emanating from pressure equipment, and comprises:

Analysis with:
- ✓ determination of operating parameters
- ✓ identification of possible hazards
- ✓ assessment of hazards

Hazard evaluation:
- ✓ adequate safety is obtained, in an iterative process by reducing the hazards.

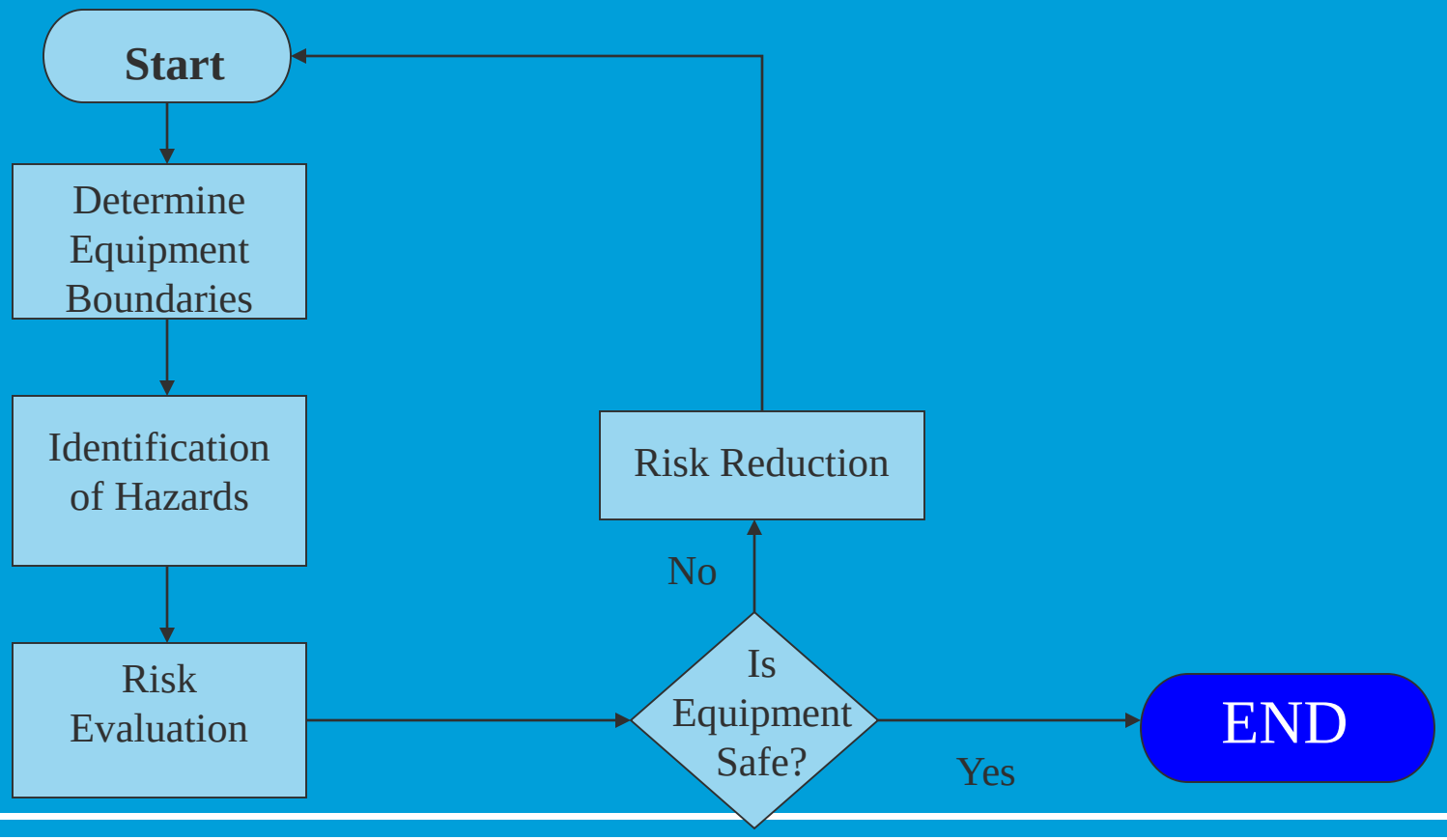
Hazard reduction by (in order of required method)
- ✓ Eliminate or reduce by technical measures
- ✓ Protection measures against hazards
- ✓ Inform users of residual hazards

Annex I - ESRs

Hazard identification should include all damage and failure mechanisms anticipated, due to operation of the pressure equipment, that may impair the integrity of the equipment, including:

- ✓ Corrosion damage in case of additional mechanical stress, such as tension crack corrosions, hydrogen-inducted crack corrosion, elongation-induced crack corrosion.
- ✓ Damage due to sheet corrosion
- ✓ Change of material properties due to ageing , brittleness, or long time rupture strength.
- ✓ Effects due to type an amount of stress, (static, pulsing, dynamic, etc.)
- ✓ Erosion
- ✓ Scheduled maintenance, opening and closing
- ✓ Clearly foreseeable misuse.

Annex I - ESRs

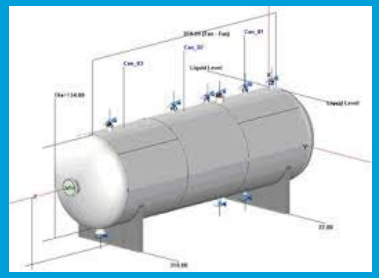

Start
Determine Equipment Boundaries
Identification of Hazards
Risk Evaluation
Is Equipment Safe?
No
Risk Reduction
Yes
END

Annex I - ESRs

General Design Requirements

Design for Adequate Strength: (loading appropriate to intended use, including):

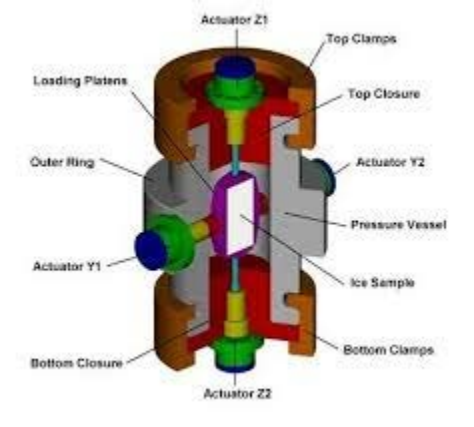
- ✓ Internal/external pressure
- ✓ Ambient and operating temperatures
- ✓ Static pressure and mass of contents in operating and test conditions
- ✓ Traffic, wind, earthquake loading
- ✓ Reaction forces and moments which result from the supports, attachments, piping, etc.
- ✓ Corrosion and erosion, fatigue, etc.
- ✓ Decomposition of unstable fluids
- ✓ Various loading which can occur at the same time must be considered



Annex I - ESRs

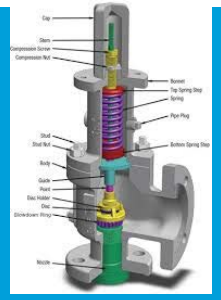
Design Requirements

- ✓ Adequate means of draining and venting (2.5)
- ✓ Adequate allowance for corrosions or chemical attack (2.6)
- ✓ Wear (2.7)
- ✓ Suitable components when used on assemblies (2.8)
- ✓ Provisions for filling and discharging (2.9)
- ✓ Protection against exceeding the allowable limits(2.10)



Annex I - ESRs

Safety Accessories



- ✓ Must be designed and constructed to be reliable and suitable for the intended duty: SIL (safety integrity level)
- ✓ Be independent of other functions
- ✓ Comply with appropriate design principles, such as, fail-safe modes, redundancy, diversity and self-diagnosis
- ✓ Pressure limiting devices must not allow the equipment to see short duration pressure surge exceeding 10% of allowable working pressure.
- ✓ Temperature monitoring devices must have an adequate response time, consistent with the measurement function.

Note: Safety Accessories are normally Category IV.
If purchased, should be CE marked by device manufacturer

Annex I - ESRs

Marking and Labelling

Marking requirements:

- ✓ Manufacturer's name, address
- ✓ Year of manufacture
- ✓ Identification according to it's nature,(type, series, serial no., etc.)
- ✓ Essential min/max allowable limits

Further information as necessary for safe installation, operation, maintenance, periodic inspection, etc.:

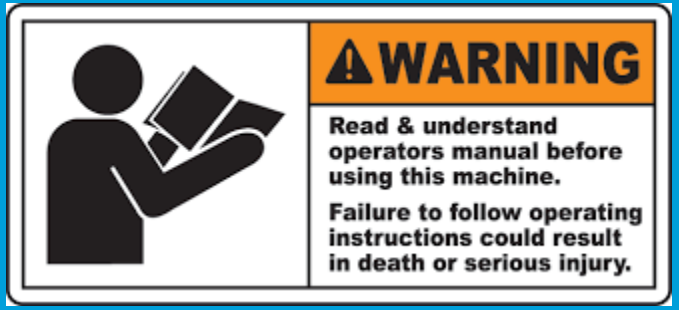
- ✓ Volume (in litre)
- ✓ Nominal size (for piping, piping accessories)
- ✓ test pressure
- ✓ Safety device set pressure
- ✓ Tare mass (in kg)
- ✓ Maximum filling mass (in kg)

Annex I - ESRs

Operating Instructions

Equipment placed on the market must be accompanied by instructions for use, including:

- ✓ Information necessary for safe, mounting/assembly of pressure equipment, putting into service, use, maintenance
- ✓ Instructions must cover information affixed to the equipment, including drawing, diagrams, etc. as need to understand the instructions
- ✓ Hazards arising from misuse, limitations, etc.
- ✓ Although not specifically mentioned in the Directive, the instructions must be in an official EU language, and in the language of intended use.



PED - Additional Requirements to ASME VIII

Hazard Analysis - must be conducted by designer

Conformity Assessment - including Notified Body involvement (except Category 1 equipment)

Welding Procedures and Welders - must be approved by a Notified Body or recognized third party for category II, III, IV. EN 287 / EN ISO 15614 or equivalent, like ASME IX +)

NDE Personnel - must be approved by recognized third party for category III, IV equipment

Materials - materials must meet European standards, or be approved by Notified Body (EAM or PMA).

Hydrostatic Testing - requirements increased from VIII, 1999 Addendum (1.43 x MAWP)

Marking - Additional marking is required

Operating instructions - required for all pressure equipment (IOM: Instruction Operating Manual)

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Conformity Assessment Modules as per 2014/68/EU

1. The references in the tables to categories of modules are the following:

I	=	Module A
II	=	Modules A2, D1, E1
III	=	Modules B (design type) + D, B (design type) + F, B (production type) + E, B (production type) + C2, H
IV	=	Modules B (production type) + D, B (production type) + F, G, H1

Conformity Assessment Modules

Internal Production Control

Module A

Manufacturer compiles technical documentation necessary for the assessment of the equipment, for conformity with the Essential Safety Requirements.
Manufacturer take all measures necessary to ensure the equipment complies with the technical documentation and Essential Safety Requirements of the Directive

Module A2

Notified Body confirms the manufacturer actually performs final assessment, through "unexpected visits", and conducts spot checks of the equipment.

Conformity Assessment Modules

Module B (Production Type) / **B** (Design type)

Manufacturer submits application to Notified Body

Manufacturer compiles and submits the technical documentation necessary for the assessment of the equipment for conformity with the Essential Safety Requirements. *For Module B(design type) this may not be based on experimental design methods of 2.2.4.*

Notified Body reviews technical documentation, assesses materials selected when using particular material appraisal, assesses welding procedures, assesses NDE personnel qualifications.

Notified Body performs or witnesses tests and examinations necessary to confirm compliance with ESR's.
Notified Body issues *EU-Type Examination* Certificate-Production type (Module B-production type) or *EU-Type Examination Certificate* (Module B-Design type)

Conformity Assessment Modules—B(Production type/design type)

Conformity to Type

Manufacturer take all measures necessary to ensure the equipment complies with the Module B(Production type)and Essential Safety Requirements of the Directive

Notified Body confirms the manufacturer actually performs final assessment, through “unexpected visits”, and conducts spot checks of the equipment.

Conformity Assessment Modules

Production Quality Assurance

Manufacturer operates an *approved* quality management system for production, final inspection and testing (generally equivalent to ISO 9002), and must ensure that the specific equipment complies with the EC Type or Design Examination Certificate (s) and the Essential Safety Requirements of the Directive.

*For Module **D1**, the manufacturer must compile technical documentation as in Module B*

Notified Body must assess the quality system through on-site audit of the manufacturer's facility, and must perform subsequent periodic audits.
Notified Body may also perform "unexpected visits" including spot checks of the equipment.

Conformity Assessment Modules

Product Quality Assurance

Manufacturer operates an *approved* quality management system for final inspection and testing (generally equivalent to ISO 9003), and must ensure that the specific equipment complies with the EC Type Certificate and the Essential Safety Requirements of the Directive.

*For Module **E1**, the manufacturer must compile technical documentation as in Module B*

Notified Body must assess the quality system through on-site audit of the manufacturer's facility, and must perform subsequent periodic audits.
Notified Body may also perform "unexpected visits" including spot checks of the equipment.

Conformity Assessment Modules

Product Verification

Manufacturer takes all measures necessary to ensure that the pressure equipment complies with the EU Type or Design Examinations and ESR's of the Directive.

Notified Body performs examination and checks on each item to verify conformity with the type or design certificates and ESR's of the directive, and in particular must:

- ✓ Verify personnel performing welding or NDE are qualified and approved (guideline 3/2)
- ✓ Verify material certificates
- ✓ Perform final inspection and check proof tests (guideline 3/6, 4/13)
- ✓ Examine safety devices
- ✓ Final documentation review (guideline 6/2)

Conformity Assessment Modules

Product Verification/UNIT verification

Module **F/G** Checklist

- ✓ Intended to be completed by design reviewer prior to fabrication inspection.
- ✓ Verify items as noted by design reviewer, or as applicable to the equipment
- ✓ Verify welding shop/department to weld-shop checklist during initial visit.
- ✓ Inspection of Assemblies includes proof testing. (guideline 3/6)
- ✓ Complete Assessment Report and attach Assessment checklist

Conformity Assessment Modules

Unit Verification (Module G)

Manufacturer submits application to Notified Body

Manufacturer compiles and submits the technical documentation necessary for the assessment of the equipment for conformity with the Essential Safety Requirements.

Notified Body *examines the design*, and performs examination and checks on each item to verify conformity and ESR's of the directive, and in particular must:

- ✓ Verify personnel performing welding or NDE are qualified and approved
- ✓ Verify material certificates
- ✓ Perform final inspection and check proof tests
- ✓ Examine safety devices

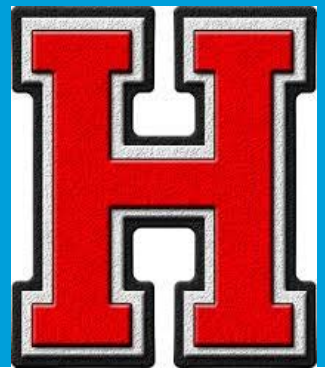
Conformity Assessment Modules

Full Quality Assurance (H Module)

Manufacturer operates an *approved* quality management system for design, production, and final inspection and testing (generally equivalent to ISO 9001), and must ensure that the specific equipment complies with the Essential Safety Requirements of the Directive.

Notified Body must assess the quality system through on-site audit of the manufacturer's facility, and must perform subsequent periodic audits *such that a full reassessment is carried out every three years*.

Notified Body may also perform "unexpected visits" including spot checks of the equipment.



Conformity Assessment Modules

Full Quality Assurance +design examination (H1 Module)

Manufacturer operates an *approved* quality management system for design, production, and final inspection and testing (generally equivalent to ISO 9001), and must ensure that the specific equipment complies with the EC Design Certificate (H1) and the Essential Safety Requirements of the Directive.

Notified Body must assess the quality system through on-site audit of the manufacturer's facility, and must perform subsequent periodic audits *such that a full reassessment is carried out every three years*.

Notified Body may also perform "unexpected visits" including spot checks of the equipment.

Conformity Assessment Modules

Full Quality Assurance (H1 Module)

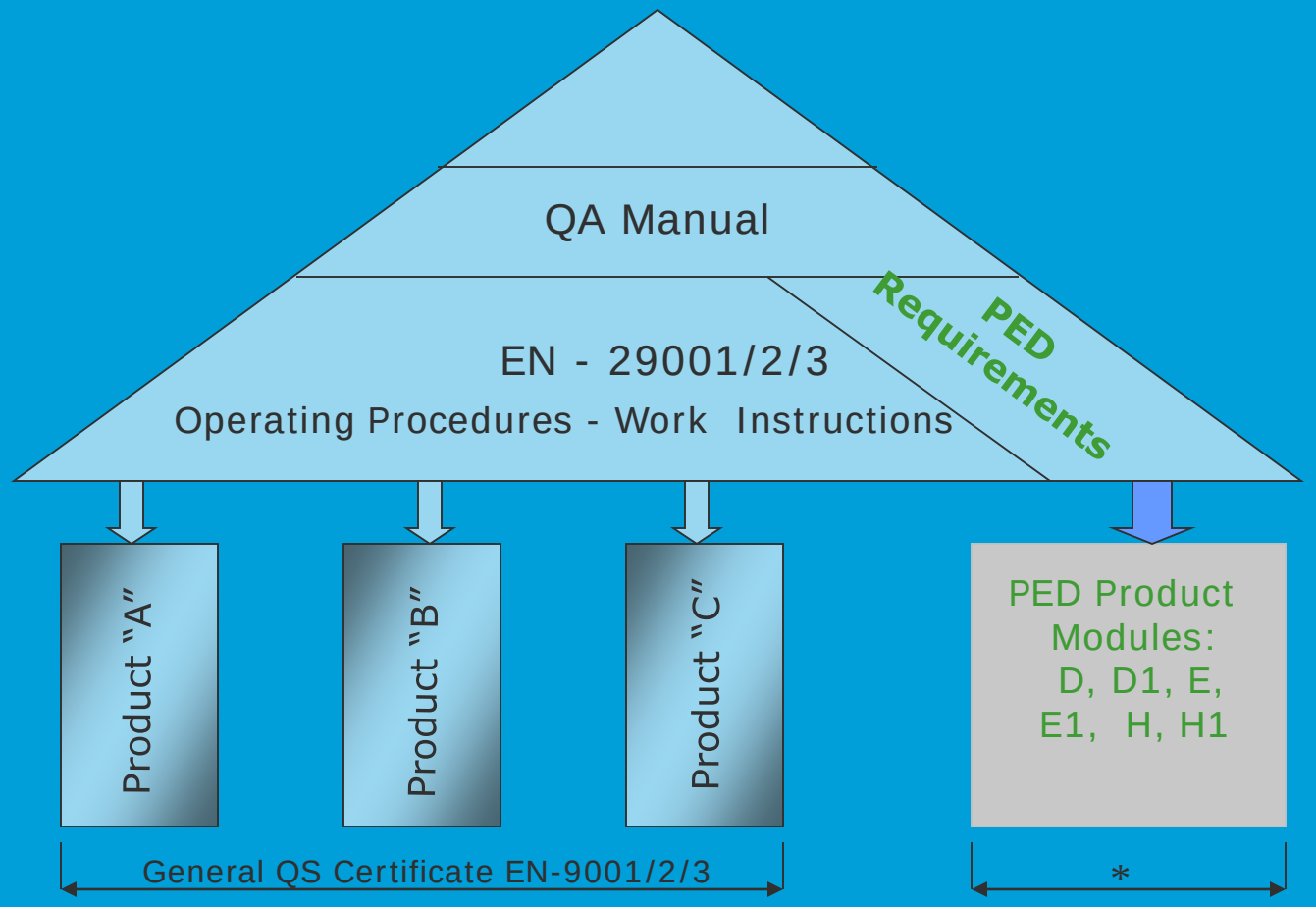
Module H1, in addition to module H:

Manufacturer compiles technical information including:

- ✓ Design specification, including standards applied
- ✓ Necessary supporting evidence, (particularly when harmonized standards are not used) including results of tests carried out

Notified Body reviews the technical information and, when provisions of the Directive are met, issues an EC Design Examination Certificate.

Conformity Assessment Modules



*** Specific Procedures for Each product. Needs Notified Body Approval**

New Approach Notified and Designated Organizations) Information System (Nando)

Lists of Notified Bodies can be searched on the NANDO web site. The lists include the identification number of each notified body as well as the tasks for which it has been notified, and are subject to regular update.

http://ec.europa.eu/growth/tools-databases/nando/

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PED in practice

Replacement fuel gas scrubber for an UK FPSO;

• Design Pressure	FV to 10 Barg
• Design Temperature	-10ºC to +80ºC
• Volume	1500 litres
• Contents	Dry Hydrocarbon Gas
• Horizontal Accelerations	+/- 0.5 m/s^2
• Vertical Accelerations	+/- 0.5 m/s^2

PED in practice

PED Assessment:

Does PED apply?	Yes, PS > 0.5 barg and equipment not ‘excluded’.
For use in European Union (EU) or European Economic Area (EEA)?	Yes, applies.

PED in practice

- What is the type of equipment?
  - a vessel?
  - piping?
  - safety accessory?
  - pressure accessory?
  - assembly?

PED in practice

- What does the equipment contain under all foreseeable operating conditions?
  - a liquid, or
  - a gas, a liquefied gas, a gas dissolved under pressure, a vapour, or a liquid with vapour pressure greater than 0.5 bar above atmospheric pressure at maximum allowable temperature.

PED in practice

What group is the contents under foreseeable operating conditions?

- Group 1? (Dangerous fluids)
  - Explosive,
  - extremely flammable,
  - highly flammable,
  - flammable (where maximum allowable temperature above flashpoint)
  - very toxic,
  - toxic,
  - oxidizing.
- Group 2?
  - All the rest!

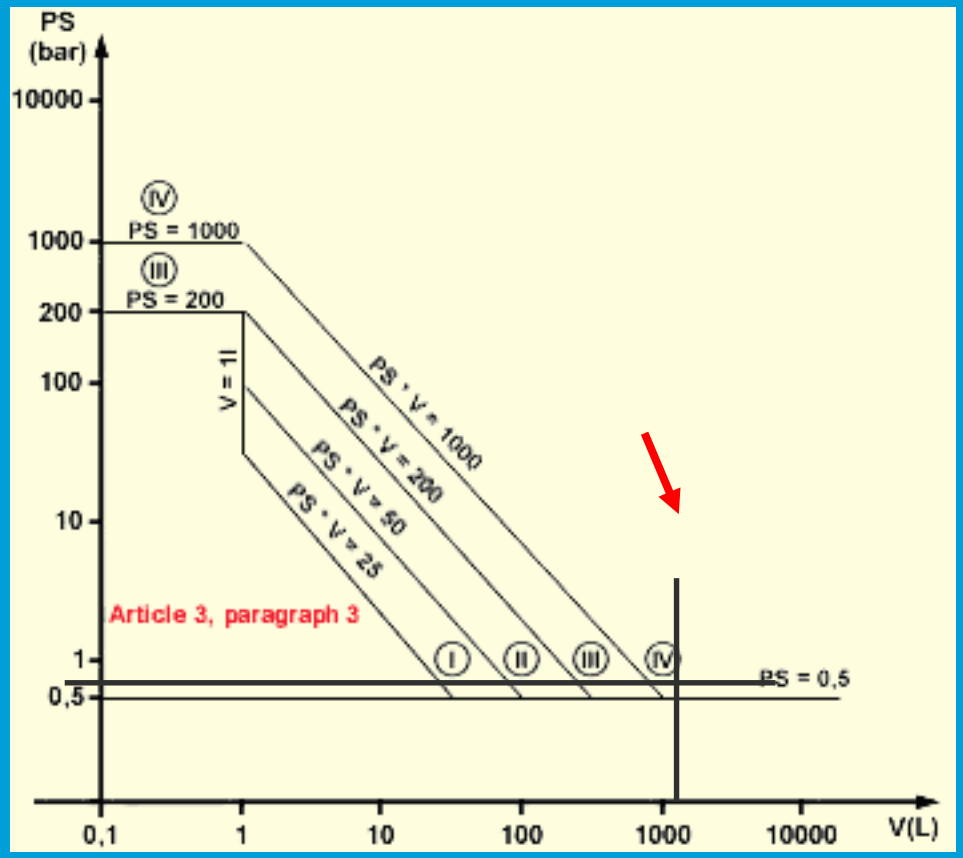
PED in practice

To find the appropriate category reference needs to be made to the graphs in Annex II of the Directive. There are 9 graphs in Annex II and Article 3 is used to find the right one.

	Vessels				Steam Generator	Piping			
State of Contents	Gas		Liquid			Gas		Liquid	
Fluid Group	1	2	1	2		1	2	1	2
Refer to Table/ Graph (Annex II of PED	1	2	3	4	5	6	7	8	9

PED in practice

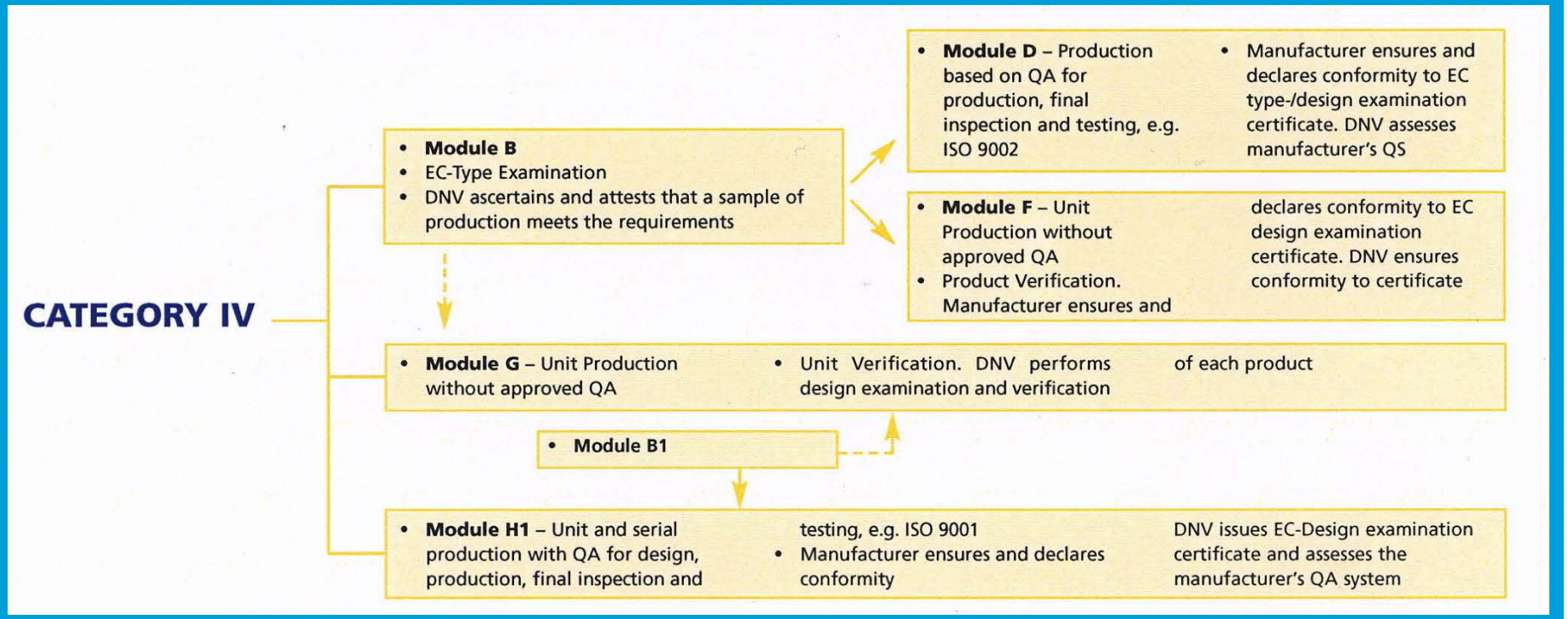
- Table 1 of Annex II:



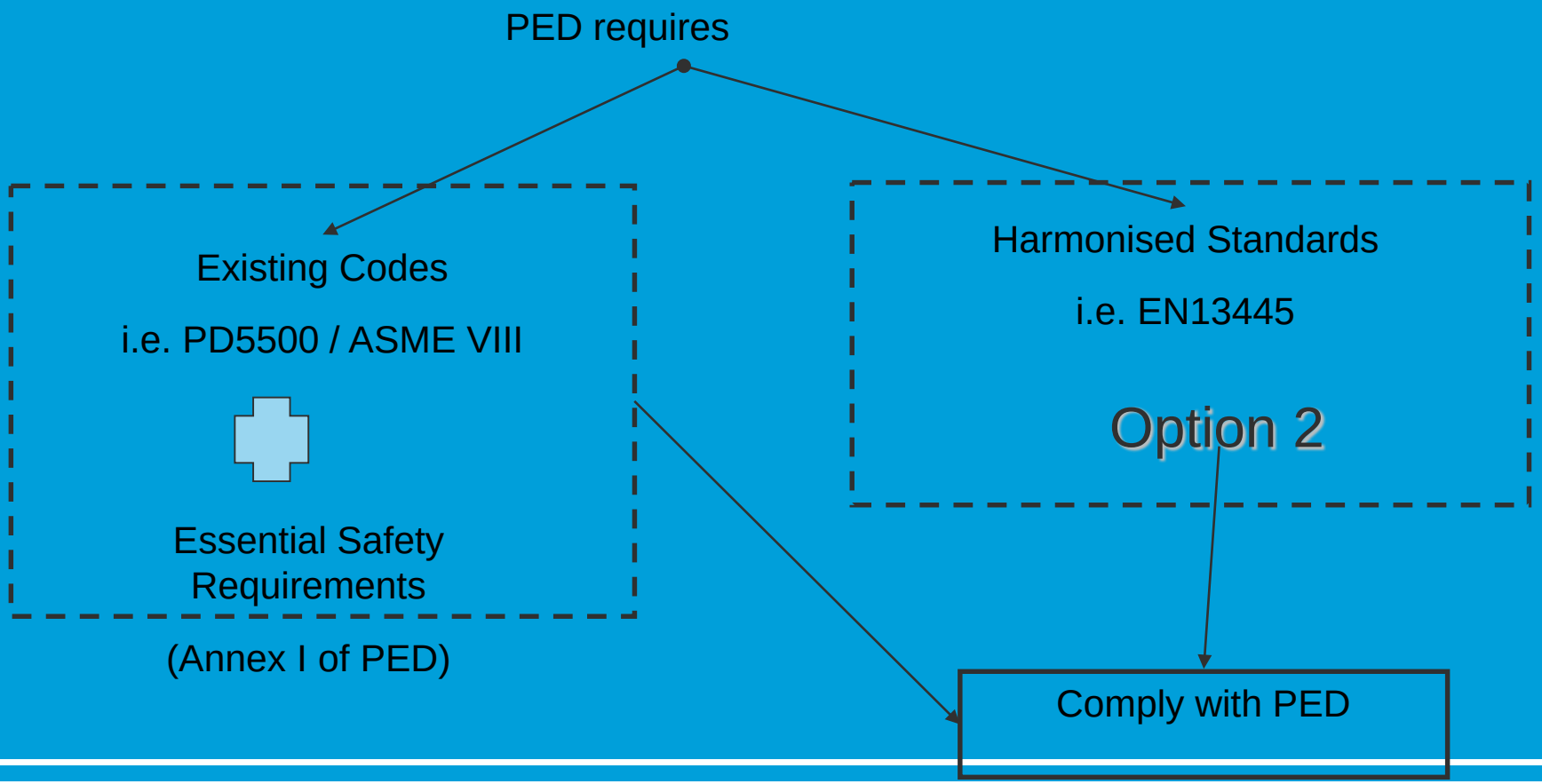
PED in practice

Next select conformity assessment module depending on:

- the equipment category,
- the manufacturers quality system,
- the product quantity,
- the manufacturer



PED in practice



PED in practice
Technical File Content (TFC) – Design Review

RISK ANALYSIS

TECHNICAL SPECIFICATION
Description of the pressure equipment
- Pressure (Highest/lowest)
- Temperature (Highest/lowest)
- Volume
- Fluid (Specification, Fluid group, Density)

CALCULATIONS
Description of the relevant load cases:
- Fatigue (Number of/allowable number of load cycles)
- Creep (Max operation time in creep range)
- Static loads
- Dynamic loads

Design Standard and Method
Special regulations
Customer Specifications

ESR-EVALUATION
To be drawn up if non harmonised standards are used

DRAWINGS
Shall contain information showing:
- All manufacturing details
- Joint efficiencies
- Marking plate

Following information can be presented on the drawings alternative on separate documentation:
- WPS
- Forming procedures
- Heat treatment
- Coating&Painting
- Brazing
- Inspection schedule
- Tests to be done
- Materials
- Marking of welds

PED in practice
TCF-content – Manufacture Control

MATERIAL CERTIFICATION
Including welding consumables

PMA
If non-harmonised materials are used

TEST REPORTS

PROOF TEST REPORT
Inclusive proof test procedure

INFORMATION SHOWING (IF NOT SHOWN ON DRAWINGS)
- WPS
- Forming procedures
- Heat treatment
- Coating&Painting
- Brazing
- Inspection schedule and record
- Tests to be done
- Materials
- Marking of welds

OPERATION MANUAL

WPS
Approved by NB/RTPO for Cat II, III and IV

LIST OF WELDERS
Approved by NB/RTPO for Cat II, III and IV

NDT-RECORDS
NDT-operators shall be approved by NB/RTPO for Cat III and IV

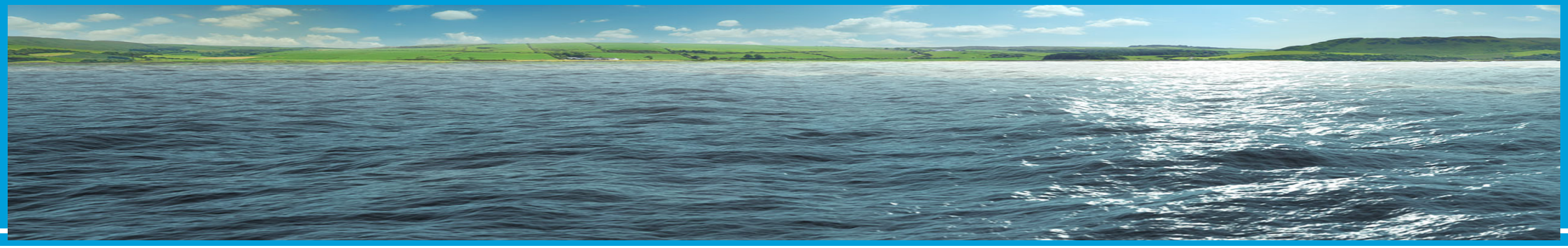
Do you know the new **PED 2014/68/EU and others**

Not only the PED changes, but also a number of other EU directives will change, due to the new NLF, like:

Directive	Products	Revision
PED 97/23/EC	Pressure Equipment	New PED (2014/68/EU) in force 19 July 2016
LD 95/16/EC	Lifts	New LD (2014/33/EU) in force 20 April 2016
R&TTE 99/5/EC	Radio Equipment	Replaced by new Radio Equipment Directive 2014/53/EC on 13 June 2016
EMC 2004/108/EC	Electromagnetic Compability	New EMC (2014/30/EU) in force 20 April 2016
LVD 2006/95/EC	Low Voltage	New LVD (2014/35/EU) in force 20 April 2016
ATEX 94/9/EC	Devices in Explosive Atmosphere	New ATEX (2014/34/EU) in force 20 April 2016
RCD 94/25/EC	Recreational Crafts	New RCD (2013/53/EU) in force 18 January 2016
MID 96/98/EU	Measurements Instruments	New MID (2014/32/EU) in force 30 October 2016

Why choose DNV GL ?

- ✓ Notified for all conformity assessment modules, plus material approval, welders
- ✓ Notified for many other directives like CPR,ATEX etc.
- ✓ International network of offices and resources
- ✓ Internationally recognized by many National Authorities
- ✓ Experience with most related industries onshore and offshore
- ✓ Extensive service range, from pre-award services through in-service inspection
- ✓ Local competent resources.
- ✓ ASME Accredited AIA services
- ✓ “One stop shop Solution “



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Thank You

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